

North Coast (Areas 3 & 4) Recreational Angling Creel Survey 2021

FINAL REPORT



Prepared for:

North Coast - Skeena First Nations Stewardship Society
Prince Rupert, BC

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EXECUTIVE SUMMARY

Due to the potential impact on multiple transboundary stocks of salmon, the North Coast (Areas 3 and 4) recreational fishery presents as a high priority for monitoring. Effort and harvest data, as well as biological samples and genetic information, are required for management of fisheries and monitoring of indicator stocks. The North Coast Skeena First Nations Stewardship Society (NCSFNSS) has operated this program since 2015, with support from the Pacific Salmon Commission Northern Fund. The survey consists of access point angler interviews and aerial overflights, typically for the months of June through August but also including May since 2019. Lodge-based recreational harvest and effort are monitored by DFO via a voluntary logbook program. Survey design and implementation, and analytical methods are similar to and consistent with those used in Area 3 and 4 creel survey monitoring since 2008.

Since 2020, the Area 3 and 4 creel survey was required to implement new safety practices and adapt the angler interview procedure due to COVID-19. Creel observers were masked, and positioned themselves at least 2 metres from the anglers during interviews. In order to prevent infection while collecting biosamples, the use of gloves, disinfectant, and masks was required. This allowed surveyors to gain some access to catch for biosample collection and species ID.

From 1 May to 31 August 2021, a total of 2,628 vessel interviews (involving 8,011 anglers) were collected by staff stationed at one of three access points (located in Prince Rupert or Port Edward) during 245 survey shifts. 32 aerial effort surveys were conducted. High priority was placed on gathering biosamples and fin-clip incidence rates from Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon. When possible, length measurements were collected from Pacific Halibut (*Hippoglossus stenolepis*). Survey effort was adequate, as relatively good precision was achieved for the harvest estimates for most of the main taxa (all precision goals were met).

Total fishing effort was estimated to be 12,252 vessel trips. In total, an estimated 17,025 Chinook Salmon were caught, with 7,383 fish kept and 9,643 released. No Chinook Salmon were harvested during the non-retention period from 19 June - 11 July 2021. It was estimated that 44,289 Coho Salmon were caught, with 38,126 fish kept and 6,163 released (this was the largest Coho Salmon harvest estimated since the creel program began in 1995). Total Pacific Halibut catch was estimated to be 24,787 individuals, with 16,671 fish kept (biomass estimated at 92.9 net tonnes) and 8,116 released. Total harvest estimates included 7,343 Lingcod (*Ophiodon elongates*), 11 Yelloweye Rockfish (*Sebastes ruberrimus*), 8,195 other rockfishes (*Sebastes* spp.), and 1,449 groundfish other than Pacific Halibut and Lingcod.

Creel surveyors biosampled 614 Chinook Salmon, collecting length measurements (median 71 cm), adipose fin-clip incidence rates (9.8% were clipped), and scale samples. For Coho Salmon, 1,164 fish were biosampled (median fork length 62 cm; 0.2% adipose clipped). Heads were collected from 48 Chinook Salmon and 13 Coho Salmon for submission to the Salmon Head Recovery Program. Biosampled Pacific Halibut (n=1,073) had a median length of 79 cm, and average net weights of 5.8 kg and 5.4 kg in Areas 3 and 4, respectively.

During the month of May, fishing effort and Chinook Salmon harvest were estimated at 1,785 boat trips, and 1,580 fish. The Department of Fisheries and Oceans Canada (DFO) expressed interest in a May creel survey because genetic samples would help them resolve stock composition and determine exploitation rates on early-run-timing stocks. During the month of May 2021, we biosampled 185 Chinook Salmon, taking 145 scale samples, and processing 15 tagged heads.

Recreational fishing effort and salmon catch in Areas 3 & 4 have been increasing over the last 27 years, in some cases dramatically. Despite the long-term trend, effort in 2020 and 2021 were below

the predicted value likely as a result of COVID-19. Despite reduced effort, Coho Salmon harvests were higher than that predicted by the long-term increasing trend. Chinook Salmon harvests were lower than expected, likely as a result of the no-retention period. Pacific Halibut harvest did not show a strong among-year trend, likely the result of effective management measures that reduced daily bag limits and imposed maximum size limits. While the increasing trends highlight the need for continued monitoring of this fishery into the future, there is evidence that management measures can keep harvest under control.

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INTRODUCTION

The North Coast recreational fishery (in DFO's Pacific Region, Management Areas 3 and 4) harvests multiple species, mainly targeting Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon, but also Pacific Halibut (*Hippoglossus stenolepis*), Lingcod (*Ophiodon elongates*), and other groundfish, rockfishes (*Sebastes* spp.), and invertebrate species. Residents and tourists both enjoy sport fishing on the North Coast, and over the past 25 years fishing effort has increased substantially (Robichaud and Addison 2021). In light of increasing pressure, and of conservation concerns for some target stocks, there is a need to understand harvest from the Area 3 and 4 recreational fishery, with respect to catch rates and stock origins. And since harvested salmon can originate from BC or US (enhanced and wild) stocks, there is a critical need for this creel program to determine fishery impacts, especially for stocks of concern.

The creel survey program helps the Pacific Salmon Commission develop improved information for resource management, including better stock assessment, data acquisition, and improved scientific understanding of factors that might limit salmon production. In addition to producing harvest and effort estimates with known error, this creel program has the added benefit of collecting direct field observations of catch, which allows for species identifications and harvest quantities to be verified, and biological samples to be collected from Chinook and Coho salmon. For example, CWT data, which are important components of the annual stock assessment data, and are required to monitor trends and manage the fisheries for Nass and Skeena river Chinook and Coho salmon stocks, come from the heads that our observers collect from randomly sampled adipose-clipped Chinook and Coho salmon.

The Area 3 and 4 creel survey program has operated during each recreational fishing season since 2008, estimating fishing effort and species-specific catch rates (Van Tongeren and Winther 2010, Robichaud and Addison 2021). Starting in 2015, the responsibility for carrying out the creel survey shifted from Fisheries and Oceans Canada (Prince Rupert) to the North Coast Skeena First Nations Stewardship Society (NCSFNSS), with support from the Pacific Salmon Commission Northern Fund. NCSFNSS's creel survey (Robichaud et al. 2016) followed a design similar to that used in the past by Fisheries and Oceans Canada, which, in turn, was based on an adaptation of the Strait of Georgia Creel Survey design, developed in 1982 by LGL Limited (English et al. 2002).

This report documents the 2021 recreational catch and effort statistics for fishing activities in Pacific Fishery Management Areas 3 and 4, and describes the program activities, data collection methodology and the analytical approaches that were used to generate the estimates.

SCOPE

The geographic scope of the creel survey was the same as in the previous twelve years (Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017, Robichaud and Addison 2018, 2019, 2020, 2021), i.e., Management Areas 3 and 4 (from Portland Inlet, through Chatham Sound to Porcher Island; Figure 1). For the purposes of data collection and analysis, the study area was

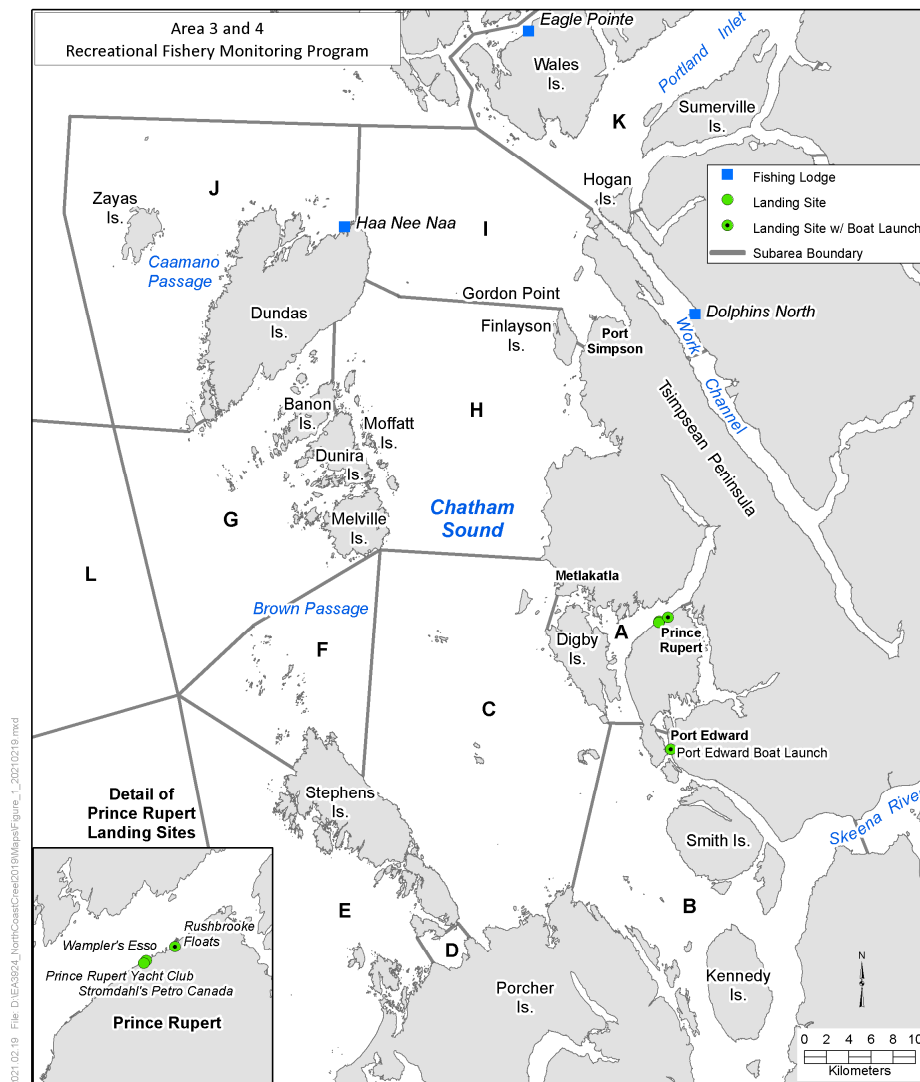


Figure 1. Map of the North Coast study area, showing the major landing sites, fishing lodges, and the boundaries of the subareas, which were used as spatial strata in the creel survey design.

divided into 12 geographic strata ('subareas'), including three subareas in Area 3 (3I-3K) and nine subareas in Area 4 (4A-4H, and 4L).

The temporal scope of the full study was from 1 May through 31 August 2021. Survey schedules were designed to provide sufficient data to derive statistically robust catch and efforts estimates for each month and subarea. However, mid-season it was decided to use five temporal strata, termed "periods", which better coincided with changes in the management measures regulating the fishery than did the biologically 'arbitrary' calendar months. For ease of among-year comparison, the period-specific data were reconstituted into monthly summaries (Table 1) wherever possible. For this reason, the third period was sometimes divided into two sub-periods (Table 1): one sub-period for the June days (19 -30 June), and the other sub-period for the July days (1-11 July).

Table 1. Delineations of temporal strata. The five Periods coincided with changes in the fishery regulations. Periods were further divided into six Sub-periods to allow data to be easily grouped into four Months.

<i>Period</i>	<i>Sub-period</i>	<i>Month</i>
1-31 May	1-31 May	May
1-18 June	1-18 June	June
19 June - 11 July	19-30 June	
	1-11 July	July
12-31 July	12-31 July	
1-31 August	1-31 August	August

The study included creel analysis of all major local sportfish species, with greater emphasis placed on Chinook and Coho salmon, Pacific Halibut and Lingcod.

GOAL AND OBJECTIVES

The goal of this study was to conduct a creel survey of the North Coast (Areas 3 and 4) recreational fishery following the design and methodology used recently by DFO (Van Tongeren and Winther 2010), for the purpose of producing estimates of effort and catch of all species targeted by the recreational fishery. The creel survey included aerial effort counts and access point angler interviews, conducted throughout Areas 3 and 4 from May through August 2021. The precision goal for the study was to estimate total harvests of large components (such as Chinook Salmon) within 15% of the true value 19 times out of 20. Creel survey strata included temporal separation by month (or period) and day type (weekdays vs. weekend days). Spatial stratifications were based on creel survey subareas. All local fishing areas (i.e., those accessed from major boat launches in the area) were included in our survey design.

The seven objectives for this study were:

1. Maintain software for data management and analysis comparable to DFO's CREST program;
2. Provide monthly (May-August) estimates of the catch and effort of all salmon species targeted by the recreational fishery and associated estimates of precision;
3. Provide DFO and PSC with monthly in-season catch estimates for all salmon species harvested in the North Coast recreational fishery;
4. Provide communication materials to recreational harvesters to convey the usefulness and importance of their catch data in fisheries management;
5. During angler interviews, collect biological data from a representative subsample of the landed harvest, including fin clip incidence, lengths, and scale or tissue samples for Coho and Chinook salmon;
6. Produce a final written report to be available to a technical audience; and
7. Present creel survey findings at the North Coast salmon post-season review meeting to a broad audience of commercial, recreational interests, and First Nations.

Objective 1 was addressed over the course of the study period, and included the need to stratify data by management period instead of by month. Objectives 2 and 6 were met by completing

this report. Objective 3 was met throughout the study period by providing timely and complete summaries of the monthly survey activities (with complete creel analyses done for each update report). Objective 4 will be met subsequent to the finalization of this report, as its contents will be based on the results presented herein. Objective 5 was met by biosampling fish at access points throughout the study period: fish were measured for length and sexed, scale or tissue samples and adipose fin clip incidence data were collected for Coho and Chinook salmon, heads were collected from adipose-clipped fish, and all biological data were submitted to DFO at the end of study period. Objective 7 was met in December 2021 at the DFO IHPC Post-Season Review meeting.

METHODS

A complete census of catch and effort was available for lodge-based anglers. However, the remainder of the recreational fishery occurred over too large an area to completely census catch and effort. Instead, statistically robust methods were used to estimate catch by multiplying estimates of *angler effort* by estimates of *catch per unit of effort*. Separate estimates were generated for each creel monitoring subarea, month, day type (weekday vs. weekend), and species.

For each creel subarea during each period and day type, fishing effort was estimated by counting boats that were actively fishing during overflight surveys (sub-area counts were reduced by the number of lodge-based boats thought to have been visible to the aircraft). Catch per unit of effort for non-lodge boats was estimated from interviews (see data forms in Appendix 1) conducted at known access points. During interviews, anglers were asked about their catch, effort, and fishing locations. They were also asked about their hourly fishing activity patterns.

Data collected during interviews included:

- Date;
- Number of anglers in party;
- Whether or not the trip was guided by a professional;
- Subarea angler activity – for each subarea, the hours during which angling activity was conducted;
- Subarea catch – the number of fish kept and the number released by species and subarea;
- Subarea gear – the amount and type of fishing gears used to catch each species in each subarea;
- Subarea targets – the number of hours spent targeting each species in each subarea (anglers that targeted only shellfish were noted);
- Biosampling – if allowed by the anglers, harvested Chinook and Coho salmon were measured (nose to fork length), sexed, checked for adipose clips and, if clipped and permission given, heads were collected for the Salmon Head Recovery Program. Scale samples were taken from Chinook Salmon, and tissue samples (fin clips) from Coho Salmon. Chinook Salmon were also examined for flesh colour. Pacific Halibut were measured (total length);
- Refusals – whether or not the anglers refused an interview, allowed catch to be observed, or allowed catch to be biosampled; and
- Weather conditions.

The analytical methods used, similar to those used in previous years (e.g., Robichaud and Addison 2021), provided a statistically unbiased estimate of catch per effort, assuming the anglers interviewed were representative of the entire non-lodge recreational fishery. Interviews were conducted at the three main access locations: Port Edward and Rushbrooke boat launches, and Prince Rupert Yacht Club. The locations surveyed were selected from all available access points, based on their geographical distribution and the amount of fishing activity that is typically conducted from that site, in order to maximize the number of interviews obtained. This approach was based on two important observations: 1) the variability in CPE (catch-per-effort) among fishing parties landing at a single access point tends to be as great as the variability in CPE among different access points within a geographic area; and 2) CPE and effort can vary substantially both within and between days at a single site (English et al. 2002). Under these conditions it is better to obtain a large number of interviews covering all temporal strata for a small number of sites than to sample a larger number of sites and obtain fewer interviews and less complete temporal coverage for any specific site.

During an observer shift, attempts were made to interview all anglers returning to the access location on that day, but some were missed (it is assumed that a random process dictated whether a boat was missed or interviewed), and some anglers refused to participate (it is not clear how the angling behaviour of the refusals differed from that of the participants). Ideally, interviews would capture data about angler activity in all creel subareas, but the statistical subarea where the anglers fished dictated the geographic distribution of our samples. It is assumed that the geographic distribution of fish sampled at the access locations was random relative to the fish harvested by the fishery.

The frequency of interview shifts was planned with the target of sampling 20% of estimated effort, while attempting to have “hands on” and direct observations of (and biological samples from) as many fish as possible meant allocating adequate time to each interview. The interview schedule was designed to capture data from representative angler parties in both day types, and over all time periods of the day. Interviewing sessions were separated into AM and PM shifts, with AM shifts occurring between 7:00 and 15:00, and PM shifts from 13:00 to 21:00. In total, 245 shifts were worked.

Fishing effort was estimated from boat-counts made during overflights. Overflights were conducted on 2-3 weekend days and 5-6 weekdays each month. Mid-season, in response to the announcement of Chinook Salmon management measures¹, we partitioned our temporal strata into five ‘periods’ (rather than the typical four months), which meant that period-specific sample sizes were lower than what they would have been had we generated *monthly* effort estimates). Timing of the overflights was targeted for peak fishing periods (11 AM to 1 PM), and only one flight (1 August 2021) occurred at an off-peak time (12:43 to 14:32). Analytical methods accommodate the occasional off-peak flight (e.g., due to weather or scheduling conflicts) which can result in increased uncertainty in the effort estimate.

¹ Salmon management measures are sometimes released in June or July, and implementation of measures and restrictions can vary among years, thus requiring us to respond *in season* with modifications to our study design.

Angler Activity Patterns

Two weighting factors were used together with the interview-derived angling activity data to estimate the daily non-lodge fishing activity pattern (English et al. 2002). The first weighting factor, $W1$, expanded the numbers of days spent interviewing at each landing site to account for the total number of days available for sampling. That is, it was assumed that the daily activity pattern recorded during the interview shifts at landing site a , were consistent for landing site a , even during the days when no interviews occurred. A specific $W1$ was calculated for each landing site during each period and day type:

$$W1_{pda} = \frac{N_{pd}}{K_{pda}}, \quad (\text{Equation 1})$$

where N_{pd} was the total number of type d days in period p ; and K_{pda} was the number of days during which interviews occurred at landing site a , on type d days during period p .

The second weighting factor, $W2$, expanded the numbers of interviews conducted, to account for the recreational fishing boats that were *not* interviewed. Some recreational fishing boats were not interviewed because of refusals, and some boats returned during periods of high traffic volume when it was not possible to interview all returning vessels at once. Some returning vessels were not doing recreational fishing at all -- they may have been taking a pleasure cruise, or conducting Food-Social-Ceremonial fishing. For the recreational fishing vessels, it was assumed that the activity pattern recorded during the interview shifts also held for those anglers that were not interviewed. A specific $W2$ was calculated for each surveyor (u) operating on each surveying date (k) at each landing site (a):

$$W2_{aku} = \frac{LF_{aku} + LM_{aku} \left(\frac{LF_{aku}}{LF_{aku} + LN_{aku}} \right)}{LF_{aku} - ri_{aku}} \quad (\text{Equation 2})$$

where LF_{aku} was the number of interviewed boats that were fishing (a subset of these, ri_{aku} , refused an interview, or had an incomplete interview that could not be used), LN_{aku} was the number of interviewed boats that were not fishing, and LM_{aku} was the number of boats that were not interviewed by surveyor u , during surveying date k , at landing site a . This equation incorporates the assumption that interviewed boats were representative, allowing us to estimate the portion of missed boats that were fishing.

We used the term $A_{pdsakut}$ to denote the number of non-lodge boats reporting activity during time-block t , as interviewed by surveyor u on survey date k , at landing site a , in subarea s , during period p , and on day type d (n_{pdsaku} was used to denote the total number of boats that reported activity during *any* time-block). The two correction factors were applied, and the data were summed over surveyors, survey dates and landing sites:

$$A'_{pdst} = \sum_k \sum_u \sum_a (W1_{pda} \cdot W2_{aku} \cdot A_{pdsakut}). \quad (\text{Equation 3})$$

Summing the adjusted number of anglers over the hourly time-blocks gave:

$$T'_{pds} = \sum_t A'_{pdst}. \quad (\text{Equation 4})$$

The proportion of non-lodge boats (P_{pdst}) that were actively fishing during each of the hourly time-blocks (t) was calculated for each period, day type, and subarea:

$$P_{pdst} = \frac{A'_{pdst}}{\sum_k \sum_u \sum_a (W1_{pda} \cdot W2_{aku} \cdot n_{pdsaku})} \quad (\text{Equation 5})$$

Table 2. The amount of data (number of complete interviews) available to estimate non-lodge angler activity patterns, for all levels of each factor, 2021.

Period	Day Type	Subarea												TOTALS		
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall
May	WD	3	1	0	53	27	43	9	2	60	17	17	0	4	228	232
	WE	1	2	7	82	63	46	16	5	63	7	13	1	10	296	306
	Total	4	3	7	135	90	89	25	7	123	24	30	1	14	524	538
1-18 June	WD	11	15	2	78	17	31	23	0	51	9	12	2	28	223	251
	WE	2	2	5	49	28	43	33	1	52	9	8	0	9	223	232
	Total	13	17	7	127	45	74	56	1	103	18	20	2	37	446	483
19 June - 11 July	WD	22	30	1	110	20	35	15	0	53	10	46	3	53	292	345
	WE	16	44	6	125	41	67	20	1	54	21	49	5	66	383	449
	Total	38	74	7	235	61	102	35	1	107	31	95	8	119	675	794
12-31 July	WD	20	63	7	144	52	73	67	5	134	12	31	6	90	524	614
	WE	7	59	12	83	48	44	40	3	96	9	21	5	78	349	427
	Total	27	122	19	227	100	117	107	8	230	21	52	11	168	873	1,041
August	WD	44	24	29	98	50	57	43	2	72	8	29	5	97	364	461
	WE	29	35	24	84	45	59	45	2	67	17	25	4	88	348	436
	Total	73	59	53	182	95	116	88	4	139	25	54	9	185	712	897
GRAND TOTAL		155	275	93	906	391	498	311	21	702	119	251	31	523	3,230	3,753

Using this method, 120 unique angler activity patterns were to be estimated (i.e., 5 periods $\times$ 2 day types $\times$ 12 subareas). To reliably describe angler activity, a relatively large number of boats (~60) needed to be interviewed in each of the 120 blocks. In the end, some blocks contained too few interviews (Table 2), so it was decided to pool activity data over subarea. The equation for non-lodge angler activity was thus:

$$P_{pdt} = \frac{\sum_s A'_{pdst}}{\sum_s \sum_k \sum_u \sum_a (W1_{pda} \cdot W2_{aku} \cdot n_{pdsaku})} \quad (\text{Equation 6})$$

and the associated variance was:

$$S^2_{P_{pdt}} = \frac{(P_{pdt})(1 - P_{pdt})}{\sum_s \sum_k \sum_u \sum_a (W1_{pda} \cdot W2_{aku} \cdot n_{pdsaku})} \quad (\text{Equation 7})$$

Catch Per Effort Estimation

Catch per effort (numbers of fish kept or released per boat trip) of non-lodge boats was estimated for each species from interviews of non-lodge anglers. Catch per effort was calculated separately for guided and non-guided boats, and overall; but only the overall estimates were used in subsequent calculations. For each interview (i), the guided status (g), period (p), and the catch (C) of each species (r) in each subarea (s) was recorded. Given that each interview represented one boat trip, then catch per effort (CPE), in terms of catch per boat trip, was calculated as:

$$CPE_{psrgi} = \frac{C_{psrgi}}{1} \quad \text{where } g \in \{\text{all, guided, not-guided}\} \quad (\text{Equation 8})$$

Mean CPE was calculated as the average catch, over all n_{psg} boat trips:

$$C\hat{P}E_{psrg} = \frac{\sum_{i=1}^{n_{psg}} C_{psrgi}}{n_{psg}} \quad (\text{Equation 9})$$

The variance for the estimate of mean catch per effort was calculated as:

$$S_{C\hat{P}E_{psrg}}^2 = \frac{\sum (CPE_{psrgi} - C\hat{P}E_{psrg})^2}{(n_{psg} - 1)} \quad (\text{Equation 10})$$

In the event that a period/subarea-specific sample size was < 5, adjacent subareas would be pooled together as follows:

- if n for 3I, 3J or 3K was too low, then it was combined with the other two subareas, along with data from ambiguous parts of Area 3 (“3?”) to give a 3?IJK regional average CPE;
- if n for 4A, 4B or 4C was too low, then it was combined with the other two to give a 4ABC regional average CPE;
- if n for 4D, 4E or 4F was too low, then it was combined with the other two to give a 4DEF regional average CPE; and
- if n for 4G, 4H or 4L was too low, then it was combined with the other two to give a 4GHL regional average CPE.

Similar methods were used to calculate the average number of fish harvested (retained) per boat trip (HPE), and the average number of fish released per boat trip (RPE).

Angler Effort Estimation

To obtain statistically valid estimates of non-lodge fishing effort, boats were counted during overflight surveys conducted from a fixed-wing aircraft traveling along a set path over the study area (see Figure 2). In order to calculate the non-lodge effort, the estimated counts of active lodge boats needed to be subtracted from the overflight counts for that particular day.

Lodge Effort Census

A complete census of lodge-based angling effort was collected from lodge log books by DFO personnel. DFO used the following procedure to estimate the number of active lodge vessels that would have been available to be observed during the flights: 1) Lodge based vessels were considered unavailable to the flights on change-over days (days when the lodges received new clients; on typical change-over days, departing clients leave before the flight and new clients arrive after the flight). Change-over days occurred twice a week but timing was often different for each lodge, making 29% (2/7) of the lodge effort not visible to flights due to change-over days. 2) Lodges reported their lunchtime activity to DFO, which ranged from virtually always going in for a mid-day break to rarely leaving the water during the day. Thus, approximately 60% of the remaining vessel activity was available to the flight counts on ‘regular’ days (i.e., not change-over). 3) Vessels on mid-day break or on change-over days were excluded from the numbers of vessels visible to the flights. All of the lodge catch was reported from Area 3, without subarea-specific effort information. Thus, lodge effort was partitioned among the Area 3 subareas, see below.

Non-lodge Effort Surveys

Table 3 shows the number of overflight surveys scheduled for each day type and period. Each survey was designed to allow visual inspection of the subareas through which the aircraft passed. Each time the aircraft entered a subarea, the start and end times were recorded, along with the

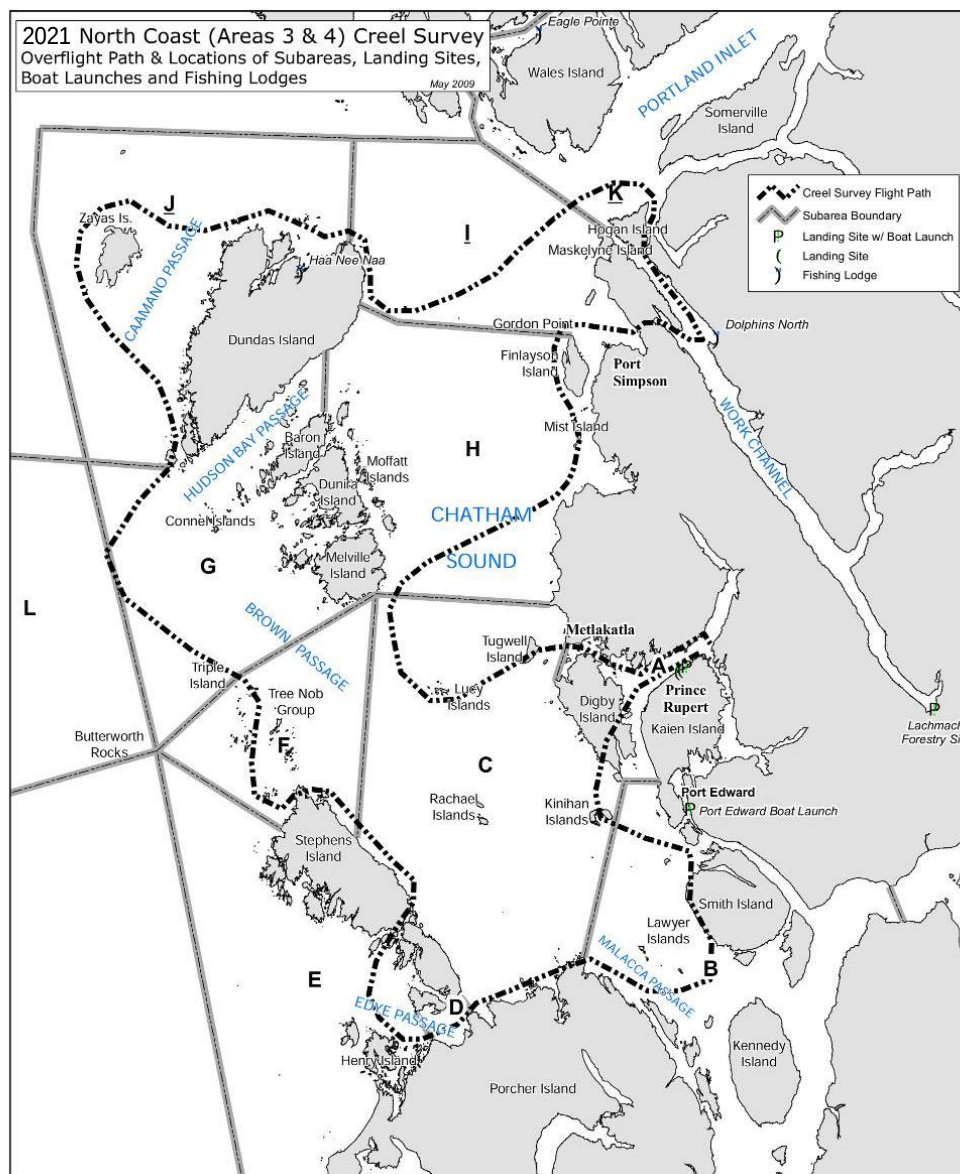


Figure 2. Map showing planned flight path of effort survey overflights. Letters indicate subareas. Flights originated and ended in Prince Rupert and flew in the clockwise direction. On good weather days, the flights also included a foray into subarea 4L (path not shown).

Table 3. Overflight sampling effort (number of flights shifts) by day type and month (left) or period (right), 2021. * Flight scheduled for July 31, meant to represent weekend effort during the 12-31 July period, was delayed until 1 August due to heavy fog.

Month	Weekday	Weekend/ Holiday	Total	Period	Weekday	Weekend/ Holiday	Total
May	5	3	8	May	5	3	8
June	5	3	8	1-18 June	3	2	5
July *	5	3	8	19 June - 11 July	4	2	6
August	5	3	8	12-31 July *	3	2	5
Total	20	12	32	August	5	3	8
				Total	20	12	32

numbers of small (< 30 ft) and large (≥ 30 ft) recreational vessels that were observed, tallied separately depending on whether or not they were actively fishing at the time.

During survey o (conducted during period p and on day type d), observers tallied the total number of recreational boats that were actively fishing at time t on their i^{th} pass over subarea s , V_{pdsoit} . For each visit to a subarea within Management Area 3, the relative proportion of boats was calculated, ρ_{pdsoit} , and the total number of lodge boats (λ_{pdo}) apportioned accordingly:

$$\lambda_{pdsoit} = \rho_{pdsoit} \cdot \lambda_{pdo} . \quad (\text{Equation 11})$$

The number of non-lodge boats was hence:

$$v_{pdsoit} = V_{pdsoit} - \lambda_{pdsoit} . \quad (\text{Equation 12})$$

Since angling occurs over the course of the entire day, the number of non-lodge boats that were observed was divided by the proportion of average daily number of boats active (P_{pdst}) during the time block when the observations were recorded. Since the flight path (Figure 2) brought the observers into some subareas more than once during the survey (covering different ground, thus the tallies were considered additive), the adjusted tallies were summed over all visits, to calculate the total number of active non-lodge fishing boats on the day of the survey, by subarea, B_{pdso} :

$$B_{pdso} = \sum_i \frac{v_{pdsoit}}{P_{pdst}} . \quad (\text{Equation 13})$$

The variance of B_{pdso} was calculated using the standard formulas for combining the variance of products and quotients of two independent random variables (Goodman 1960):

$$\begin{aligned} \text{if } z = x/y, \text{Var}(z) &= (y^{-2})\text{Var}(x) + (x^2y^{-4})\text{Var}(y) ; \\ \text{if } z = xy, \text{Var}(z) &= (y^2)\text{Var}(x) + (x^2)\text{Var}(y) ; \text{ and} \\ \text{if } z = qx, \text{Var}(z) &= (q^2)\text{Var}(x) \text{ where } q \text{ is a constant.} \end{aligned} \quad (\text{Equation 14})$$

Since there was no variance associated with the boat counts, it followed that

$$S_{B_{pdso}}^2 = (v_{pdsoit})^2 (P_{pdst})^{-4} S_{P_{pdst}}^2 . \quad (\text{Equation 15})$$

Next, the survey-specific ‘daily effort’ estimates were averaged over the number of surveys conducted, n_{pds} , as:

$$\hat{B}_{pds} = \frac{\sum_{o=1}^{n_{pds}} B_{pdso}}{n_{pds}} \quad (\text{Equation 16})$$

with a variance of:

$$S_{\hat{B}_{pds}}^2 = (n_{pds})^{-2} S_{B_{pdso}}^2 . \quad (\text{Equation 17})$$

Total fishing effort for each sub-period was calculated for each subarea by multiplying the period-specific average daily effort (e.g., $\hat{B}$ for the ‘19 June – 11 July’ period was applied to both the ‘19-30 June’ and ‘1-11 July’ sub-periods, see Table 1) by the number days of day type d that occurred in sub-period c :

$$E_{cds} = \hat{B}_{pds} \cdot N_{cd} , \quad (\text{Equation 18})$$

which was then summed over day type to give:

$$E_{cs} = \sum_d \hat{B}_{pds} \cdot N_{cd} , \quad (\text{Equation 19})$$

with a variance of:

$$S_{E_{cs}}^2 = \sum_d S_{\hat{B}_{pds}}^2 \cdot N_{cd}^2 . \quad (\text{Equation 20})$$

Total monthly fishing effort was calculated by summing over the appropriate sub-periods (i.e., total effort for July was the sum of effort in the ‘1-11 July’ and ‘12-31 July’ sub-periods). As always, variances for sums were additive (Goodman 1960).

Because of inclement weather, not all overflights surveyed every subarea. Especially for the more exposed subareas, it was possible to have no overflight data in a given combination of period and day type. In this event, subarea-specific effort was scaled using: 1) the proportion of effort in that subarea, relative to that in a three-subarea cluster (see pooling of adjacent subareas for CPE estimation, above), measured during overflights for which all three subareas were surveyed; and 2) the effort in the cluster's other two subareas, estimated for the combination of period and day type during which the data gap occurred.

Catch and Harvest Estimation

Total catch was calculated for each sub-period, subarea, and species by multiplying sub-period-specific angling effort by period-specific catch per effort, and then summing over day type:

$$C_{csr} = \sum_d (E_{cds} \cdot C\hat{P}E_{psrg}) \quad (\text{Equation 21})$$

where g = 'all' (see Equation 8). The standard errors for these catch estimates were:

$$S_{C_{csr}} = \sqrt{\sum_d \left(E_{cds}^2 \frac{S_{CPEpsrg}^2}{n_{psg}} + CPE_{psrg}^2 \frac{S_{E_{cds}}^2}{n_{pds}} + \frac{S_{CPEpsrg}^2}{n_{psg}} \frac{S_{E_{cds}}^2}{n_{pds}} \right)} \quad (\text{Equation 22})$$

where n_{pds} and n_{psg} are the sample sizes for E and CPE estimates, respectively. Total monthly catch was calculated by summing over the appropriate sub-periods, with additive variances.

Similar methods were used to estimate monthly numbers of fish harvested, and monthly numbers of fish released by replacing CPE with HPE or with RPE in Equations 21 and 22.

Biosampling

When given permission by the anglers, surveyors collected biological data from Chinook and Coho salmon and Pacific Halibut. Biosampled species were sexed when they were intact. Chinook and Coho salmon were examined for clipped adipose fins (mark rate data used by DFO for coded wire tag analyses) and when possible, heads from clipped fish were collected. The flesh color was noted for Chinook Salmon. Scale samples were collected from Chinook Salmon, and tissue samples (fin clips) were collected from Coho Salmon (used by DFO to determine age, and to carry out genetic stock identification).

Ideally, all anglers would allow their catch to be biosampled, and the examined fish would represent an unbiased sample of all harvested fish. But Robichaud et al. (2016) theorized that the interview protocol, used until the end of 2015, may have resulted in bias. The protocol had surveyors ask anglers about adipose clips before asking permission to observe the fish. Anglers were often engaged in a conversation about the purpose of the fin-clips and how the data would be used, and it appeared that anglers with clipped fish were more likely to permit handling and biosampling of catch than when no clipped fish were present. To avoid this bias, we modified our data collection protocol in 2016 so that anglers were asked if their catch could be biosampled before any other interactions took place. Also starting 2016, no fin clip data were accepted on the word of the anglers: the only fin-clip data we used were those which came from observers when they handled and inspected the catch.

Biomass of Harvested Pacific Halibut

Halibut lengths were converted to weights using the International Pacific Halibut Commission (Clark 1992) regression equations:

$$\begin{aligned}\text{Weight(lb)}_{\text{net}} &= 6.921 \times 10^{-6} \times \text{FL(cm)}^{3.24} \\ \text{Weight(lb)}_{\text{round}} &= 9.166 \times 10^{-6} \times \text{FL(cm)}^{3.24}\end{aligned}\quad (\text{Equation 23})$$

Converting average length to average weight allowed harvested biomass to be calculated from harvested fish counts.

Statistical Analyses

For each fish taxa, differences in CPE among periods were examined using Kruskal Wallis tests (Zar 1984), whereas differences between trip-types (guided vs. non-guided) were examined using paired t-tests (i.e., the difference between guided vs unguided catch for a given subarea and period was tested against a null of zero; Sokal and Rohlf 1995). Kruskal Wallis tests were used to examine the effects of period and day type on effort, and the effects of month on catch and harvest. When statistically significant effects were observed, *post-hoc* Kruskal-Wallis tests were used to compare between each possible pair, where the false discovery rate was controlled using the Benjamini–Hochberg procedure (Benjamini and Hochberg 1995).

Trailer Census

Starting at the beginning of May, the total number of parked trailers at the landing locations was recorded daily during the peak of the activity timing window (i.e., between noon and 2 PM; similar to the targeted timing of aerial overflights). Theoretically, trailer counts could be used as an index of fishing effort, are more economical than overflights, and are not weather dependant, though no sub-area-specific effort data are collected. DFO has been collecting trailer count data for years and we are maintaining the time series. Linear regression was used to determine the relationship between trailer counts and fishing effort. A data point was generated each time a trailer census occurred on the same day as an overflight. For the regression analysis, the location-specific trailer counts and non-lodge boat counts were used as the independent and dependent variables, respectively.

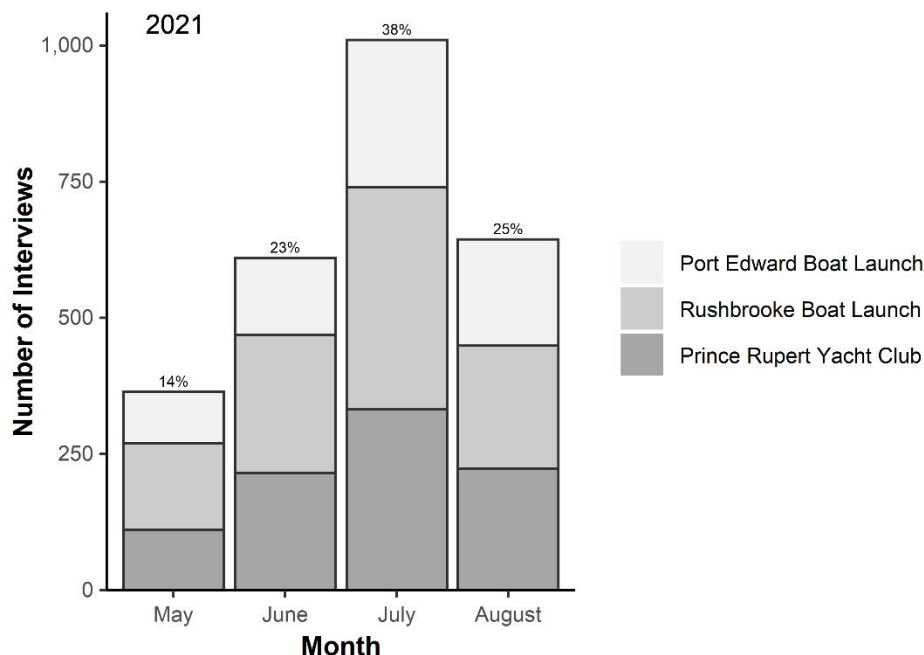


Figure 3. Number of interviews conducted at each of three landing sites, by month, 2021.

RESULTS

Angler Interviews

Over the four month study period, there were 245 observer shifts, during which 2,628 interviews were conducted, involving 8,011 non-lodge anglers (Figure 3; Table 4). July was the busiest month, with 1,010 interviews conducted (average 15.8 interviews per shift), which represented 38% of the total number of study period interviews (Table 4).

Non-lodge boat trip activity was unevenly distributed among the subareas, with 4F and 4A being the favourites (Table 5). Subareas 4C, 4B, and 4D were the next most likely areas for anglers to have reported activity. The subareas with the lowest number of trips reporting activity were 4E and 4L. Only 112 interviews were of parties that only targeted shellfish, and those were overwhelmingly in subarea 4A (Table 6).

On average, 24% of the non-lodge boat trips were guided by a professional. The guiding rate varied by landing site. Only 14 of 701 trips (2.0%) interviewed at Port Edward were guided, whereas trips departing from Rushbrooke and Prince Rupert Yacht Club were guided 14% and 53% of the time, respectively (Table 7).

In all there were 12 refusals (Table 5), or 0.5% refusal rate. The highest refusal rate (0.8%) was at Rushbrooke Boat Launch, where 67% of the refusals occurred. The refusal rates for the Port Edward Boat Launch and the Prince Rupert Yacht Club were 0% (no refusals) and 0.5% (33% of the refusals), respectively.

Table 4. Monthly number of shifts, and numbers of boats interviewed per landing site, 2021. Non-lodge effort is expressed in units of boat trips. Sampling efficiency is also calculated.

Landing Site	May				June				Jul			
	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint
Port Edward Boat Launch	94	227	20	4.7	142	373	20	7.1	270	767	23	11.7
Prince Rupert Yacht Club	111	334	20	5.6	215	723	20	10.8	332	1192	19	17.5
Rushbrooke Boat Launch	159	393	20	8.0	253	715	20	12.7	408	1258	22	18.5
TOTAL	364	954	60	6.1	610	1,811	60	10.2	1,010	3,217	64	15.8
% of Total Interviews			14%				23%				38%	
Estimated Non-lodge Effort			1,713				2,251				3,933	
% of Non-lodge Effort			15%				20%				35%	
% of Effort Sampled			21%				27%				26%	

Landing Site	August				TOTAL			
	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint
Port Edward Boat Launch	195	579	20	9.8	701	1946	83	8.4
Prince Rupert Yacht Club	223	791	21	10.6	881	3040	80	11.0
Rushbrooke Boat Launch	226	659	20	11.3	1046	3025	82	12.8
TOTAL	644	2,029	61	10.6	2,628	8,011	245	10.7
% of Total Interviews			25%					
Estimated Non-lodge Effort			3,315				11,212	
% of Non-lodge Effort			30%					
% of Effort Sampled			19%				23%	

Table 5. Fishing locations as reported by interviewed non-lodge parties, by month and subarea, 2021. For interviews that reported activity in multiple subareas, the boat trip tally was evenly partitioned among the subareas visited.

	Subarea															
Month	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	3?	4?	Refusal	TOTAL
May	3	2	3	82	71	63	15	5	84	14	20	0			1	364
June	16	29	10	140	60	104	55	1	120	25	43	5			4	610
July	38	118	16	183	91	120	105	6	216	25	78	7		2	6	1,010
August	46	43	40	100	76	91	70	4	115	17	36	6			1	644
TOTAL	103	192	69	505	298	378	245	15	534	81	177	17	0	2	12	2,628

Table 6. Numbers of non-lodge boat trips that reported only fishing for shellfish, by month and subarea, 2021.

Month	Subarea														TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	3?	4?	
May	0	0	0	18	0	1	0	0	1	1	4	0	0	0	25
June	0	0	0	26	3	0	0	0	0	0	0	0	0	0	29
July	0	0	1	27	2	1	0	0	0	0	0	0	0	0	31
August	1	0	0	23	1	1	0	0	0	0	2	0	0	0	27
TOTAL	1	0	1	94	6	3	0	0	1	1	6	0	0	0	112

Table 7. Numbers of guided (i.e., led by professional guides) and unguided boat trips, by landing site, 2021.

Landing Site	Trip Type		% guided
	Guided	Unguided	
Port Edward Boat Launch	14	687	2.0%
Prince Rupert Yacht Club	465	416	53%
Rushbrooke Boat Launch	142	904	14%
TOTAL	621	2,007	24%

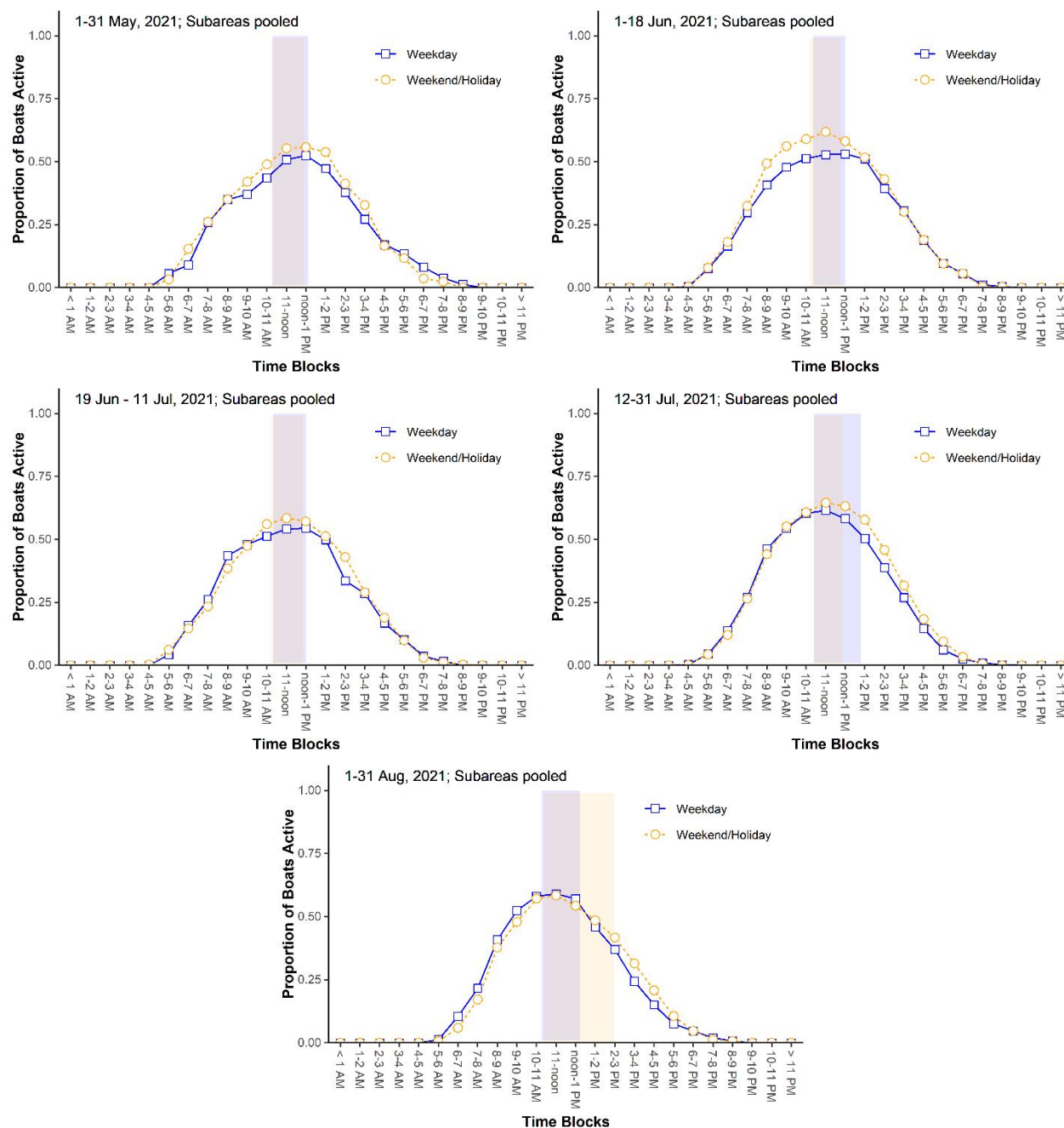


Figure 4. Angler activity pattern (proportion recreational angling boats that were actively fishing during each hour-long time block) by day type and period, pooled over all subareas, 2021. Shaded regions bound the earliest and latest flight times for each day type (blue = weekday; orange = weekend/holiday) and period.

Angler Activity Patterns

Non-lodge angler activity pattern estimates varied little among periods (Figure 4). There was virtually no difference between the weekday and weekend/holiday activity patterns, except for a slightly greater tendency for weekend anglers to be active during the peak times or in the late afternoon, as compared to weekday anglers. Timing of our overflights lined up with peak activity levels for all but one flight (1 August 2021) that occurred at an off-peak time (12:43 to 14:32) as a result of fog earlier in the day.

Catch Per Effort Estimates

Sample sizes were adequate ($n \geq 5$) to produce reliable estimates of non-lodge CPE in most combinations of periods and creel study sub-areas, (Table 8) except in subareas 3I and 3J in May, in subarea 4L in May and early June, and in subarea 4E in June, early July, and August. However, when the interview data were split between guided and non-guided groups, period-specific sample sizes did not suffice for guided trips for several more combinations of period and management area (Table 8). In those cases where sample sizes were limited, data from three adjacent subareas were pooled together for CPE calculations (Table 9).

Average number of fish harvested per non-lodge boat trip (HPE) was calculated for each species by period, subarea, and by trip type (Appendix 2).

HPE varied significantly among periods (Figure 5, Table 10) for Chinook ($\chi^2 = 36.8$, $df = 4$, $P < 0.0001$), Coho ($\chi^2 = 40.8$, $df = 4$, $P < 0.0001$) and Pink salmon (*Oncorhynchus gorbuscha*, $\chi^2 = 31.8$, $df = 4$, $P < 0.0001$). Chinook Salmon HPE increased from May into the start of June, but was zero during the non-retention period (19 June - 11 July). After the fishery re-opened, HPE was higher in late July than it was in August. Among period statistical comparisons yielded complex results (Table 10; Figure 5). For Coho and Pink salmon, median HPE increased over the course of the study period (Table 10; Figure 5). HPE for Rock Sole (*Lepidopsetta bilineata*) also varied significantly among periods ($\chi^2 = 12.4$, $df = 4$, $P = 0.015$; Table 10) but the among period differences were not statistically significant after controlling for the rate of false positives. There were no significant differences among periods for HPE of Chum (*O. keta*) or Sockeye salmon (*O. nerka*), or for any other species examined (including Pacific Halibut and Lingcod; Figure 5).

Guided trips had higher HPE than non-guided trips, and the differences were statistically significant for a variety of species (Table 11). In particular, guided trips harvested 0.5 more Chinook Salmon ($t_{47} = 4.9$, $P < 0.0001$), and 2.3 more Coho Salmon ($t_{47} = 5.9$, $P < 0.0001$) than unguided trips. In addition, guided trips harvested 0.6 more Lingcod ($t_{47} = 5.8$, $P < 0.0001$), 0.1 more Pacific Cod (*Gadus macrocephalus*, $t_{47} = 3.2$, $P = 0.003$), and 1.0 more Pacific Halibut per trip ($t_{47} = 5.9$, $P < 0.0001$) than unguided trips. Guided trips also harvested significantly more Black (*Sebastes melanops*) and Quillback (*S. maliger*) rockfish than unguided trips.

For lodge boat trips, HPE (and RPE for Chinook Salmon) values are shown in Table 12.

Table 8. Sample sizes for CPE (harvest per effort) estimates by period, for guided and not guided non-lodge boat trips, 2021.

Subarea													
Period	Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
May	Guided	2	0	1	38	3	11	1	0	35	6	8	0
	Unguided	2	3	6	97	87	78	24	7	88	18	22	1
	Combined	4	3	7	135	90	89	25	7	123	24	30	1
1-18 June	Guided	2	11	1	68	4	14	24	0	33	6	6	2
	Unguided	11	6	6	59	39	60	32	1	70	12	14	0
	Combined	13	17	7	127	43	74	56	1	103	18	20	2
19 June - 11 July	Guided	16	51	1	114	2	6	7	0	25	8	31	5
	Unguided	22	23	6	121	59	96	28	1	82	23	64	3
	Combined	38	74	7	235	61	102	35	1	107	31	95	8
12-31 July	Guided	6	76	1	101	18	11	17	1	36	7	13	11
	Unguided	21	46	18	126	82	106	90	7	194	14	39	0
	Combined	27	122	19	227	100	117	107	8	230	21	52	11
August	Guided	46	35	24	82	7	7	5	0	30	7	21	6
	Unguided	27	24	29	100	88	109	83	4	109	18	33	3
	Combined	73	59	53	182	95	116	88	4	139	25	54	9

Table 9. Subarea data (pooled over several adjacent areas, in some cases) used to calculate CPE in each period and subarea, for guided and not guided non-lodge boat trips, 2021.

		Subarea											
Period	Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
May	Guided	3?,I,J,K	3?,I,J,K	3?,I,J,K	4A	4A,B,C	4C	4D,E,F	4D,E,F	4F	4G	4H	4G,H,L
	Unguided	3?,I,J,K	3?,I,J,K	3K	4A	4B	4C	4D	4E	4F	4G	4H	4G,H,L
	Combined	3?,I,J,K	3?,I,J,K	3K	4A	4B	4C	4D	4E	4F	4G	4H	4G,H,L
1-18 June	Guided	3?,I,J,K	3J	3?,I,J,K	4A	4A,B,C	4C	4D	4D,E,F	4F	4G	4H	4G,H,L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4G,H,L
	Combined	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4G,H,L
19 June - 11 July	Guided	3I	3J	3?,I,J,K	4A	4A,B,C	4C	4D	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4G,H,L
	Combined	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L
12-31 July	Guided	3I	3J	3?,I,J,K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4G,H,L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
August	Guided	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4G,H,L
	Combined	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L

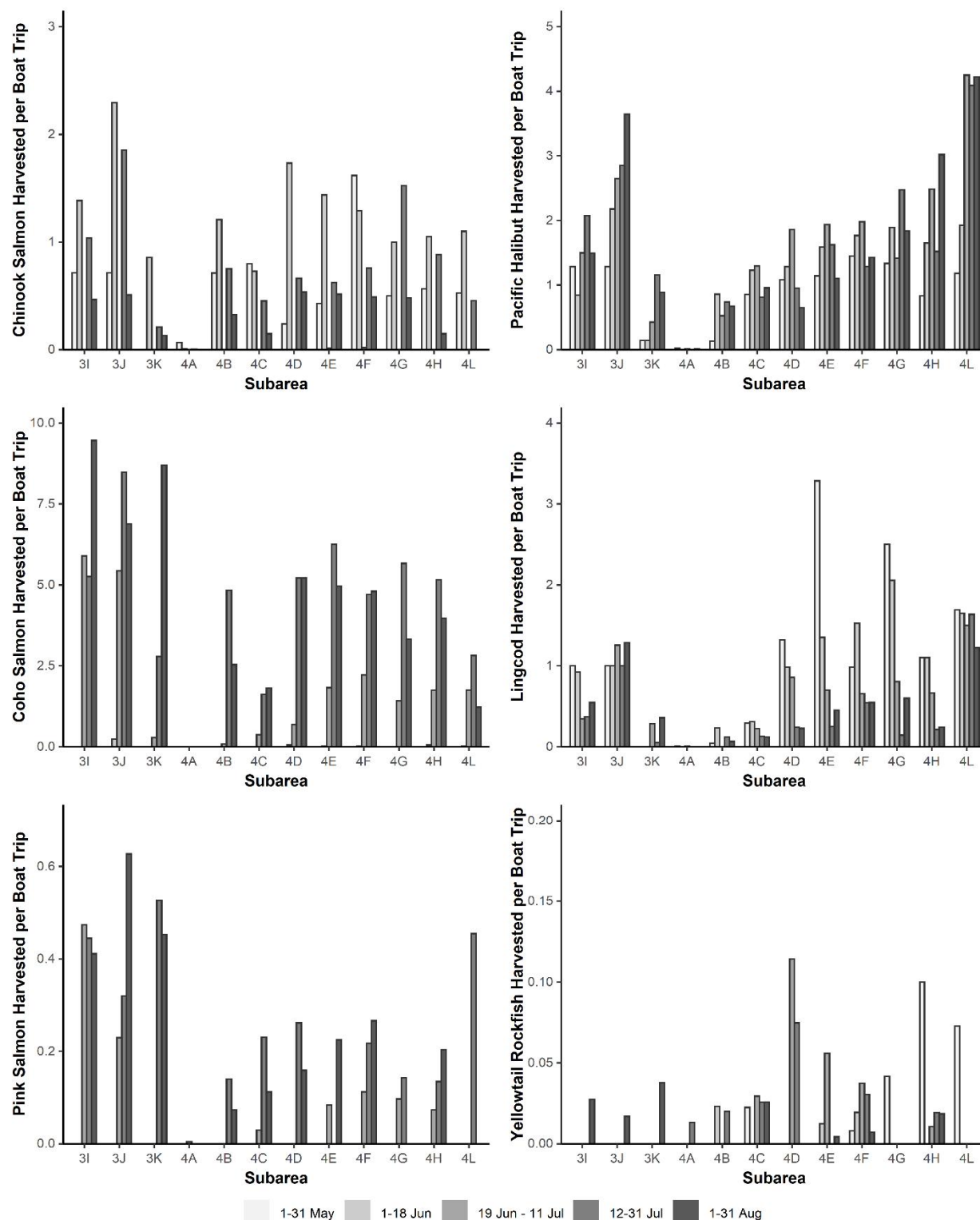


Figure 5. Average number of fish harvested per non-lodge boat trip, for six sportfish species, by period and subarea, 2021. Data are for guided and unguided trips combined. Note that Y axis scale varies among plots.

Table 10. Kruskal-Wallis tests for differences in HPE (fish harvested per non-lodge boat trip) among periods, by species, 2021. Subareas treated as replicates. Red superscript letters indicate statical differences within species (periods not connected by a common letter were statistically different).

Species	Kruskal-Wallis Statistics			Median HPE				
	χ^2	df	P	May	1-18 June	19 June-11 July	12-31 July	August
Chinook Salmon	36.8	4	<0.0001	0.5 ^{CD}	1.2 ^F	0 ^A	0.7 ^{DEF}	0.4 ^{BC}
Chum Salmon	8.1	4	0.09	0	0	0	0	0
Coho Salmon	40.8	4	<0.0001	0 ^A	0.005 ^B	1.6 ^C	5.0 ^D	4.4 ^D
Pink Salmon	31.8	4	<0.0001	0 ^A	0 ^A	0.05 ^B	0.2 ^C	0.2 ^{BC}
Sockeye Salmon	9.2	4	0.06	0	0	0	0	0
Arrow Flounder	8.6	4	0.07	0	0	0	0	0
Big Skate	3.1	4	0.55	0	0	0	0	0
Cabezon	6.4	4	0.17	0	0	0	0	0
English Sole	0.0	4	1	0	0	0	0	0
Greenling	8.2	4	0.08	0	0	0	0.004	0
Lingcod	6.5	4	0.17	1.0	1.0	0.7	0.2	0.4
Pacific Cod	3.2	4	0.53	0.03	0.08	0.1	0.07	0.1
Pacific Halibut	5.4	4	0.25	1.1	1.4	1.7	1.4	1.3
Ratfish	0.0	4	1	0	0	0	0	0
Rock Sole	12.4	4	0.015	0	0	0	0	0
Sablefish	3.2	4	0.53	0	0	0	0	0
Starry Flounder	8.9	4	0.06	0	0	0	0	0
Black Rockfish	0.9	4	0.93	0.2	0.3	0.2	0.3	0.3
Boccacio Rockfish	3.9	4	0.42	0	0	0	0	0
Canary Rockfish	5.7	4	0.23	0	0	0	0	0
China Rockfish	1.5	4	0.82	0	0	0.02	0	0
Copper Rockfish	4.1	4	0.39	0	0.02	0	0	0.006
Dusky Rockfish	3.7	4	0.45	0	0	0	0	0
Quillback Rockfish	2.4	4	0.66	0.1	0.2	0.2	0.1	0.2
Redband Rockfish	4.0	4	0.41	0	0	0	0	0
Redstripe Rockfish	4.0	4	0.41	0	0	0	0	0
Tiger Rockfish	0.3	4	0.99	0	0	0	0.005	0
Widow Rockfish	0.9	4	0.93	0	0	0	0	0
Yelloweye Rockfish	6.3	4	0.18	0	0	0	0	0
Yellowtail Rockfish	2.6	4	0.62	0	0	0	0.007	0.006

Table 11. Paired t-tests for differences in HPE (fish caught per non-lodge boat trip) between guided and non-guided boats, by species, 2021. May (when too few guided trips were recorded) data are excluded. Periods and subareas treated as replicates.

Species	Paired t-test Statistics			Median HPE		Paired Mean Difference
	t	df	P	Guided	Not	
Chinook Salmon	4.9	47	<0.0001	0.7	0.4	0.5
Chum Salmon	-1.7	47	0.10	0	0	-0.0021
Coho Salmon	5.9	47	<0.0001	4.0	1.5	2.3
Pink Salmon	1.7	47	0.09	0.02	0.05	0.05
Sockeye Salmon	0.5	47	0.59	0	0	0.002
Arrow Flounder	-2.2	47	0.035	0	0	-0.0012
Big Skate	-1.0	47	0.32	0	0	-0.0007
Cabezon	-0.9	47	0.40	0	0	-0.0019
English Sole	0.0	47	1	0	0	0
Greenling	0.7	47	0.48	0	0	0.004
Lingcod	5.8	47	<0.0001	0.8	0.3	0.6
Pacific Cod	3.2	47	0.003	0.1	0.06	0.1
Pacific Halibut	5.9	47	<0.0001	2.1	1.2	1.0
Ratfish	0.0	47	1	0	0	0
Rock Sole	-1.8	47	0.09	0	0	-0.0007
Sablefish	-1.8	47	0.07	0	0	-0.0090
Starry Flounder	-0.7	47	0.48	0	0	-0.0006
Black Rockfish	4.2	47	0.0001	0.5	0.1	0.3
Boccacio Rockfish	0.9	47	0.39	0	0	0.001
Canary Rockfish	-1.6	47	0.12	0	0	-0.0034
China Rockfish	-2.0	47	0.049	0	0	-0.0194
Copper Rockfish	0.5	47	0.61	0	0	0.005
Dusky Rockfish	-1.4	47	0.18	0	0	-0.0027
Quillback Rockfish	4.1	47	0.0002	0.3	0.1	0.2
Redband Rockfish	-1.0	47	0.32	0	0	-0.0030
Redstripe Rockfish	1.0	47	0.32	0	0	0.003
Tiger Rockfish	1.5	47	0.14	0	0	0.01
Widow Rockfish	-2.2	47	0.031	0	0	-0.0028
Yelloweye Rockfish	0.6	47	0.58	0	0	0.0008
Yellowtail Rockfish	0.1	47	0.95	0	0	0.0007

Table 12. Average number of harvested fish (and for Chinook Salmon, released) per boat trip, by month, from the census of Area 3 lodges, 2021.

Species	May	June	July	August	Total
Chinook Salmon:					
retained	1.7	0.5	0.5	0.09	0.4
released	0.2	0.7	0.4	0.03	0.3
Coho Salmon	0	1.3	3.2	4.2	2.8
Pink Salmon	0	0.004	0.010	0.01	0.009
Chum Salmon	0	0	0	0	0
Sockeye Salmon	0	0	0	0	0
Pacific Halibut	0.2	0.8	0.8	0.9	0.8
Lingcod	0.4	1.1	1.1	0.9	1.0
Other groundfish	0	0	0	0	0
Yelloweye Rockfish	0	0	0	0	0
Other rockfish	0	0.1	1.4	0	0.6

Angler Effort Estimates

Over the four month study period, a total of 32 effort surveys were conducted, including 20 weekday surveys and 12 conducted on weekends/holidays (Table 3, Appendix 3). The numbers of lodge boats that were visible during the overflights (Appendix 3) were partitioned among the subareas in Management Area 3, and subtracted from the overflight counts of small boats. The remaining small boat counts (“the non-lodge small boat counts”) were significantly correlated to trailer counts at both survey launch sites (Rushbrooke: $F_{1,30} = 192.0$, $P < 0.0001$; Port Edwards: $F_{1,30} = 99.3$, $P < 0.0001$), and when the two launch sites were combined ($F_{1,30} = 229.3$, $P < 0.0001$). The fit of the Port Edward regression (Port Edward $R^2 = 0.76$) was lower than that of the Rushbrooke regression ($R^2 = 0.86$) or that of combined-site regression ($R^2 = 0.88$; Figure 6).

The non-lodge boat counts were adjusted for non-lodge angler effort patterns in order to estimate the total monthly non-lodge angling effort for each subarea in 2021 (Table 13, Figures 7 and 8). Median daily non-lodge effort on weekdays (4.4 trips per day per subarea) was significantly lower than on weekends/holidays (6.4 trips per day per subarea; $\chi^2 = 5.2$, $df = 1$, $P = 0.022$). Effort varied among periods, ranging from 2.6 trips per day per subarea in May to 7.6 trips per

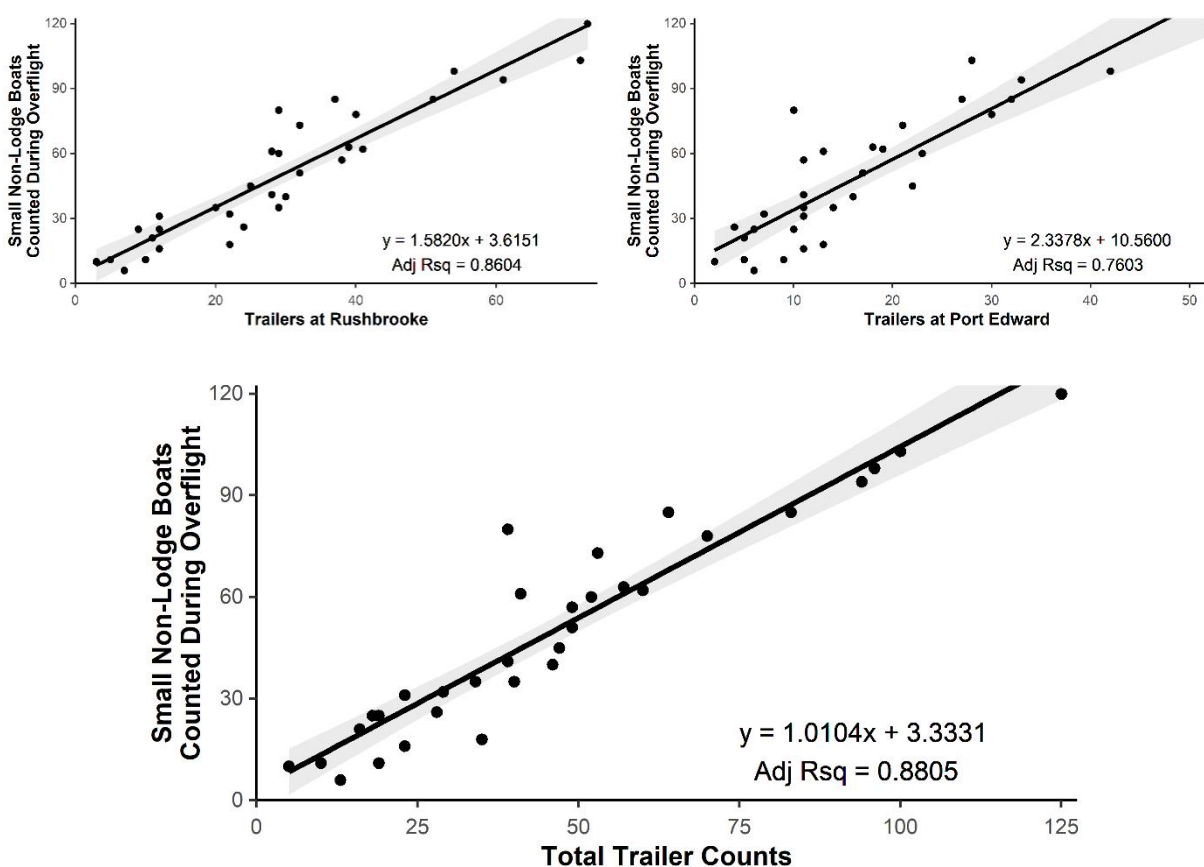


Figure 6. Linear regressions showing the relationships between small non-lodge boats counted during overflights and trailers concurrently counted at landing sites. Shaded area shows the 95% confidence interval of the predicted values. Note that X axis scale varies among plots.

Table 13. Lodge, non-lodge, and total effort estimates (boat trips per month), by month and subarea, 2021.

		Subarea												TOTALS			SE of the Totals		
Month		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May	Creel Estimate	78.9	141.2	82.8	116.6	321.1	315.1	55.5	19.9	446.4	58.0	57.9	19.9	303	1,410	1,713	5	13	14
	Lodge Report	18.7	33.6	19.7										72	0	72			
	TOTAL MAY	97.6	174.8	102.5	116.6	321.1	315.1	55.5	19.9	446.4	58.0	57.9	19.9	375	1,410	1,785	5	13	14
	SE	2.0	3.6	2.2	2.1	6.6	4.4	1.2	0.5	9.4	1.3	1.4	0.6						
June	Creel Estimate	227.2	382.4	137.1	62.0	115.9	385.7	191.3	9.7	484.6	167.2	62.9	24.7	747	1,504	2,251	7	11	12
	Lodge Report	77.9	131.1	47.0										256	0	256			
	TOTAL JUNE	305.1	513.5	184.1	62.0	115.9	385.7	191.3	9.7	484.6	167.2	62.9	24.7	1,003	1,504	2,507	7	11	12
	SE	3.0	5.5	2.3	0.9	2.5	3.8	3.4	0.5	8.0	3.5	1.1	0.6						
July	Creel Estimate	287.1	687.3	178.4	111.7	225.6	733.5	335.0	8.1	967.2	256.9	71.2	70.9	1,153	2,780	3,933	8	16	18
	Lodge Report	95.9	229.5	59.6										385	0	385			
	TOTAL JULY	383.0	916.8	237.9	111.7	225.6	733.5	335.0	8.1	967.2	256.9	71.2	70.9	1,538	2,780	4,318	8	16	18
	SE	3.1	7.6	2.2	1.2	3.3	5.7	5.0	0.2	13.3	4.6	1.0	1.1						
August	Creel Estimate	267.2	279.9	416.0	100.0	180.2	645.1	362.1	63.9	582.8	299.5	71.1	47.4	963	2,352	3,315	8	13	15
	Lodge Report	90.7	95.0	141.2										327	0	327			
	TOTAL AUGUST	357.9	375.0	557.2	100.0	180.2	645.1	362.1	63.9	582.8	299.5	71.1	47.4	1,290	2,352	3,642	8	13	15
	SE	3.4	4.2	5.2	1.4	2.2	6.6	4.9	1.0	8.5	5.2	1.1	1.2						
FULL SEASON TOTAL		1,144	1,980	1,082	390	843	2,079	944	102	2,481	782	263	163	4,205	8,046	12,252	14	27	30
FULL SEASON SE		6	11	6	3	8	10	8	1	20	8	2	2						

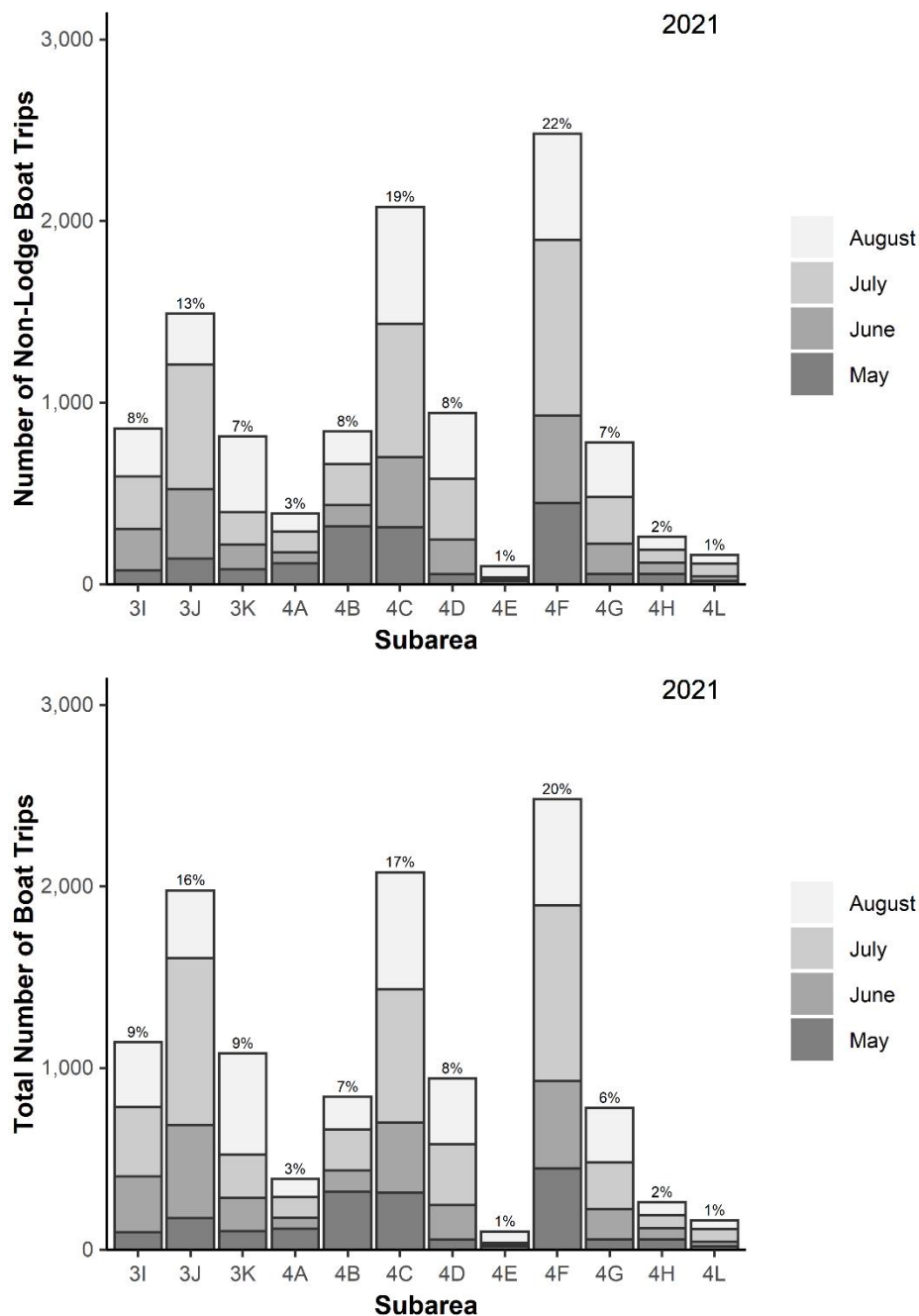


Figure 7. Total estimated numbers of boat trips, by month and subarea, 2021. Upper panel: estimated monthly non-lodge boat trips. Lower panel: total effort, calculated as the sum of lodge and non-lodge boat trips.

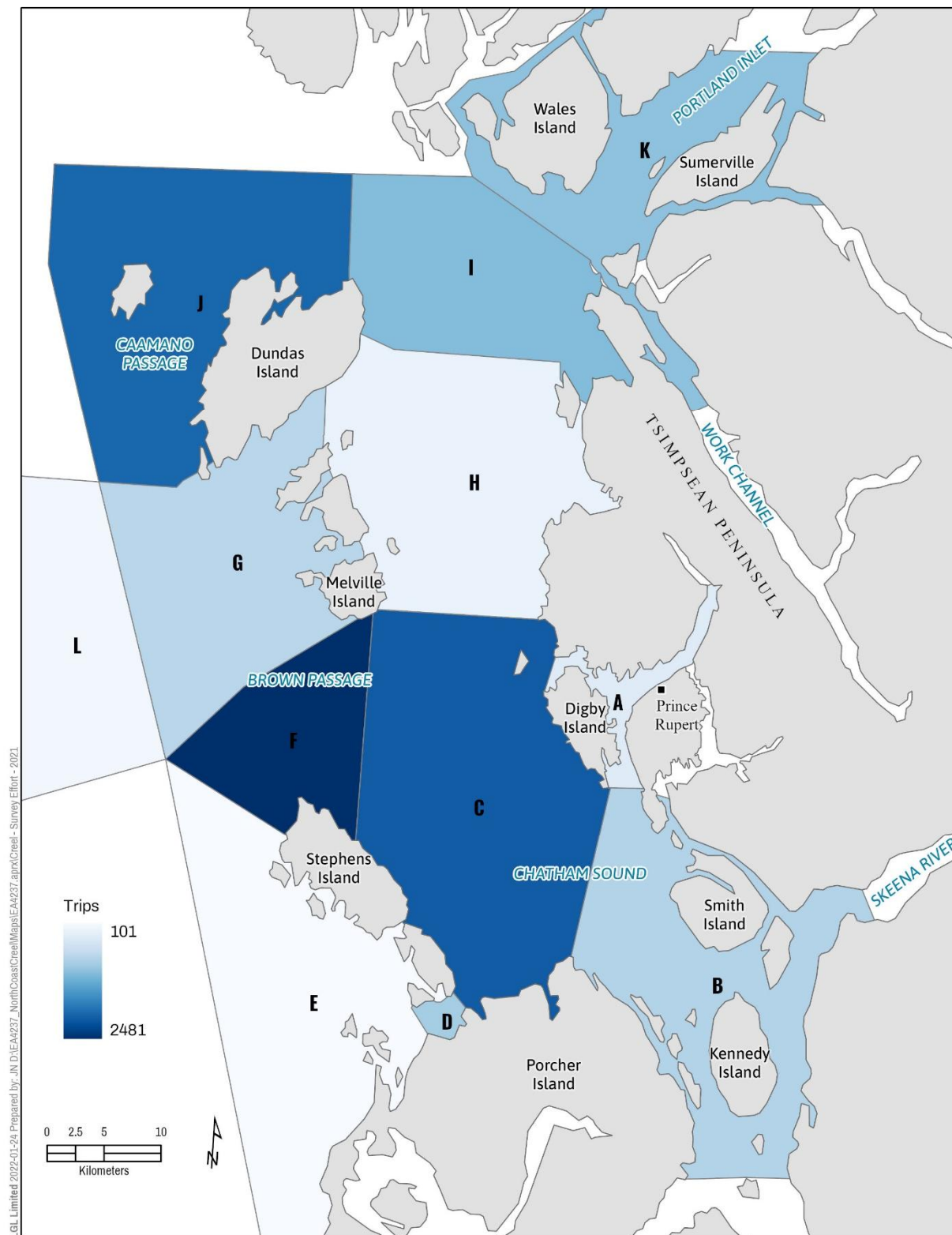


Figure 8. Heat map showing the estimated total effort (numbers of boat trips, lodge + non-lodge) in each subarea in 2021. See Table 13 for details.

day per subarea in late July, though the differences were not statistically significant ($\chi^2 = 8.3$, $df = 4$, $P = 0.080$).

Total lodge effort in 2021 was 1,040 boat trips (72 in May, 256 in June, 385 in July, and 327 in August; Table 13). To calculate total (lodge + non-lodge) fishing effort by subarea, we distributed the lodge boat trips among the three subareas in Management Area 3, according to the relative non-lodge effort in each (Table 13; Figure 7). In all, there were 1,785 (SE = 14) boat trips in May, 2,507 (SE = 12) boat trips in June, 4,318 (SE = 18) boat trips in July, and 3,642 (SE = 15) boat trips in August. The full-season (May-August) Area 3/4 effort estimate for 2021 was 12,252 boat trips (SE = 30). For comparison with our previous study years, which did not include surveys in May, the June-August effort total was 10,466 boat trips (SE = 27).

Catch Estimates

Monthly and full-season estimates were made of the numbers of fish released and retained, by species (Tables 14-23; Appendix 4). Non-lodge estimates were generated from the product of effort and CPE estimates. Lodge catches were extracted from census documents. The results presented in this section are total (lodge + non-lodge) numbers of fish retained and released.

An estimated 7,383 (SE = 215) Chinook Salmon were retained by anglers in Areas 3 or 4 during the 2021 study period (Table 15; Figure 9). The largest proportional harvest came from subareas 4F (28%) and 3J (26%). Harvest differed markedly over the study period, though the among-month differences were not statistically significant ($\chi^2 = 1.9$, $df = 3$, $P = 0.59$; Table 24). During the study, the number of released Chinook Salmon was estimated at 9,643 (SE = 402), or 57% of the total catch. Percent retention (calculated for non-lodge anglers) was 75% in May and ranged from 33-43% over the other three study months (Appendix 4).

The salmonid that was harvested in the greatest numbers during the 2021 study period was, by far, Coho Salmon (38,126 fish retained, SE = 633; Table 16; Figure 10). The largest proportional harvest came from subarea 3J (25%), followed by subareas 4F (19%), 3I (15%), and 3K (12%). Harvest differed significantly among months: harvest in May was significantly lower than that in June, which was in turn significantly lower than in July or August ($\chi^2 = 29.9$, $df = 3$, $P < 0.0001$; Table 24). During the study period, an estimated 6,163 (SE = 365) Coho Salmon were released by anglers (14% of the total catch).

Pink salmon was the least likely salmonid to be retained in 2021 (Table 17). We estimate that 8,964 (SE = 415) fish were caught, of which 21% were retained (1,859 fish, SE = 121), and 79% were released (7,105 fish, SE = 397). Harvest of Pink Salmon varied significantly among months ($\chi^2 = 22.6$, $df = 3$, $P = 0.0001$; Table 24): harvest in May was significantly lower than that in any other month; and that in June was significantly lower than in July.

Very few Chum (10 fish, SE = 4) or Sockeye (22 fish, SE = 9) salmon were harvested in Area 3 and 4 during the study period (Tables 18 and 19). A small number of harvested fish (21 fish, SE = 10) were recorded as unknown salmon species.

Table 14. Total (lodge + non-lodge) estimated harvest, by subarea, for popular sportfish taxa, 2021.

Species	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
Chinook Salmon	532	1,902	207	8	552	726	652	60	2,054	571	78	39	2,641	4,742	7,383	125	175	215
Coho Salmon	5,889	9,660	4,726	0	1,432	2,169	3,465	368	7,325	2,246	581	265	20,275	17,851	38,126	453	442	633
Pink Salmon	317	426	267	0	41	206	133	14	367	39	25	22	1,011	848	1,859	101	68	121
Chum Salmon	8	0	0	0	0	0	0	0	0	0	2	0	8	2	10	4	1	4
Sockeye Salmon	0	4	0	0	0	0	12	1	0	0	0	5	4	18	22	3	8	9
Unknown Salmon	0	0	0	0	4	5	12	1	0	0	0	0	0	21	21	0	10	10
Pacific Halibut	1,484	4,602	680	4	418	2,068	927	122	3,715	1,482	561	607	6,766	9,905	16,671	244	263	359
Lingcod	638	2,446	257	1	72	387	448	109	1,902	677	164	242	3,341	4,003	7,343	196	197	277
other groundfish *	116	215	62	0	67	226	131	27	374	70	122	41	392	1,057	1,449	60	78	99
Rockfish †	480	2,337	198	9	288	867	758	92	2,311	480	185	201	3,015	5,191	8,206	165	201	260

* includes Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and unknown groundfish

† includes Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail and unknown rockfish

Table 15. Total (lodge + non-lodge) estimated Chinook Salmon catch, by month and subarea, shown separately for released and retained fish, 2021.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	99.0	177.2	0	7.8	228.3	251.4	13.3	8.5	722.3	29.0	32.8	10.5	276	1,304	1,580	66	75	100
Released	11.4	20.4	0	0.9	153.4	60.2	4.4	0	239.5	0	5.8	1.1	32	465	497	8	49	49
June																		
Retained	125.0	519.8	111.2	0.1	114.1	141.3	253.8	13.9	425.6	109.5	12.8	7.1	756	1,078	1,834	68	59	90
Released	832.1	1098.0	87.6	0.1	77.7	173.8	258.8	9.3	710.3	203.5	160.1	40.3	2,018	1,634	3,652	187	111	217
July																		
Retained	171.3	1049.1	35.5	0.3	150.8	239.1	191.7	5.1	621.0	289.0	22.2	21.8	1,256	1,541	2,797	73	128	147
Released	672.5	1067.1	271.4	0	72.4	207.8	226.4	7.1	829.6	150.7	149.4	39.9	2,011	1,683	3,694	239	121	268
August																		
Retained	136.4	156.1	60.2	0	58.8	94.5	193.4	32.9	285.1	143.7	10.5	0	353	819	1,172	36	73	81
Released	173.4	105.2	830.9	0	49.3	116.8	144.0	19.1	138.4	215.6	6.6	0	1,110	690	1,799	153	129	200
TOTAL																		
Retained	532	1,902	207	8	552	726	652	60	2,054	571	78	39	2,641	4,742	7,383	125	175	215
Released	1,689	2,291	1,190	1	353	559	634	36	1,918	570	322	81	5,170	4,473	9,643	340	214	402
Catch	2,221	4,193	1,397	9	905	1,285	1,286	96	3,972	1,141	400	121	7,811	9,215	17,025	362	277	456

Table 16. Total (lodge + non-lodge) estimated Coho Salmon catch, by month and subarea, shown separately for released and retained fish, 2021.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
June																		
Retained	1057.4	1303.5	10.4	0	1.8	71.5	38.6	0.2	351.7	81.9	89.1	32.1	2,371	667	3,038	150	58	161
Released	96.9	106.6	0	0	0	3.8	9.1	0.2	50.0	31.7	12.8	0	203	108	311	56	27	62
July																		
Retained	1873.5	6103.5	484.7	0.3	973.2	929.3	1538.1	50.7	4176.5	1170.2	209.9	175.3	8,462	9,224	17,685	301	294	420
Released	344.1	639.1	30.6	0	158.9	202.5	357.7	1.0	715.6	316.8	47.3	0	1,014	1,800	2,814	139	174	223
August																		
Retained	2957.8	2252.6	4231.2	0	457.3	1167.8	1888.7	316.8	2796.6	994.2	281.6	57.9	9,442	7,961	17,403	303	325	445
Released	241.6	256.2	1467.7	0	74.0	222.4	288.0	41.7	318.7	95.8	31.6	0	1,966	1,072	3,038	251	129	283
TOTAL																		
Retained	5,889	9,660	4,726	0	1,432	2,169	3,465	368	7,325	2,246	581	265	20,275	17,851	38,126	453	442	633
Released	683	1,002	1,498	0	233	429	655	43	1,084	444	92	0	3,183	2,980	6,163	293	218	365
Catch	6,571	10,661	6,225	0	1,665	2,597	4,120	411	8,409	2,691	672	265	23,457	20,831	44,289	539	493	731

Table 17. Total (lodge + non-lodge) estimated Pink Salmon catch, by month and subarea, shown separately for released and retained fish, 2021.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
June																		
Retained	73.3	45.7	0	0.2	0	5.6	0	0	17.6	5.6	3.7	0	119	33	152	20	9	22
Released	56.5	106.6	8.9	0	0.4	5.6	0	0	7.4	0	1.1	0	172	14	186	52	5	52
July																		
Retained	133.0	203.5	77.1	0.2	28.2	127.9	75.6	0	194.5	33.6	6.8	21.8	414	488	902	74	50	90
Released	221.8	1421.5	200.5	0	153.2	82.7	218.7	5.1	454.7	225.8	27.0	65.5	1,844	1,233	3,076	250	120	278
August																		
Retained	110.8	177.1	190.1	0	13.3	72.3	57.6	14.4	155.1	0	14.5	0	478	327	805	65	44	79
Released	640.6	327.4	1020.3	0	19.0	328.1	201.6	65.8	792.5	395.3	51.3	0	1,988	1,854	3,842	215	178	280
TOTAL																		
Retained	317	426	267	0	41	206	133	14	367	39	25	22	1,011	848	1,859	101	68	121
Released	919	1,855	1,230	0	173	416	420	71	1,254	621	79	66	4,004	3,101	7,105	334	215	397
Catch	1,236	2,282	1,497	0	214	622	554	85	1,622	660	104	87	5,015	3,949	8,964	349	226	415

Table 18. Total (lodge + non-lodge) estimated Chum Salmon catch, by month and subarea, shown separately for released and retained fish, 2021.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
June																		
Retained	4.0	0	0	0	0	0	0	0	0	0	1.1	0	4	1	5	3	1	3
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
July																		
Retained	3.8	0	0	0	0	0	0	0	0	0	1.0	0	4	1	5	3	1	3
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
August																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	4.7	0	0	0	0	0	0	0	0	0	0	5	0	5	3	0	3
TOTAL																		
Retained	8	0	0	0	0	0	0	0	0	0	2	0	8	2	10	4	1	4
Released	0	5	0	0	0	0	0	0	0	0	0	0	5	0	5	3	0	3
Catch	8	5	0	0	0	0	0	0	0	0	2	0	13	2	15	5	1	5

Table 19. Total (lodge + non-lodge) estimated Sockeye Salmon catch, by month and subarea, shown separately for released and retained fish, 2021.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
June																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
July																		
Retained	0	4.0	0	0	0	0	0	0	0	0	0	0	4	0	4	3	0	3
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
August																		
Retained	0	0	0	0	0	0	12.3	0.8	0	0	0	5.3	0	18	18	0	8	8
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL																		
Retained	0	4	0	0	0	0	12	1	0	0	0	5	4	18	22	3	8	9
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Catch	0	4	0	0	0	0	12	1	0	0	0	5	4	18	22	3	8	9

Table 20. Total (lodge + non-lodge) estimated Pacific Halibut catch, by month and subarea, shown separately for released and retained fish, 2021.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	105.5	188.9	12.3	2.6	42.8	269.1	59.9	22.7	646.1	77.3	48.2	23.5	307	1,192	1,499	80	81	114
Released	236.6	423.6	0	0	42.8	49.6	6.7	2.8	344.8	38.7	38.6	13.0	660	537	1,197	333	84	343
June																		
Retained	339.0	1072.4	33.0	0.4	92.5	486.6	271.5	15.4	890.1	288.7	146.0	90.0	1,444	2,281	3,725	107	112	155
Released	159.3	454.5	0	0	21.5	117.7	74.1	4.9	370.1	220.7	86.9	52.1	614	948	1,562	109	105	152
July																		
Retained	573.5	2149.4	204.2	0.4	161.6	694.8	361.0	13.2	1348.8	565.1	152.7	293.5	2,927	3,591	6,518	163	157	226
Released	181.1	1270.3	129.9	0	147.2	294.0	98.1	5.1	490.7	245.0	92.2	234.6	1,581	1,607	3,188	244	160	292
August																		
Retained	465.9	1191.2	430.8	1.1	121.4	617.3	234.6	70.5	830.2	551.0	214.5	200.0	2,088	2,841	4,929	124	159	201
Released	237.9	906.2	188.4	0	24.7	144.6	78.2	20.7	234.8	179.7	64.5	89.5	1,333	837	2,169	189	129	229
TOTAL																		
Retained	1,484	4,602	680	4	418	2,068	927	122	3,715	1,482	561	607	6,766	9,905	16,671	244	263	359
Released	815	3,055	318	0	236	606	257	34	1,440	684	282	389	4,188	3,928	8,116	467	246	528
Catch	2,299	7,657	999	4	655	2,674	1,184	155	5,156	2,166	844	996	10,954	13,833	24,787	527	360	638

Table 21. Total (lodge + non-lodge) estimated Lingcod catch, by month and subarea, shown separately for released and retained fish, 2021.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	90.0	161.1	0	0.9	14.3	92.1	73.2	65.3	439.2	144.9	63.7	33.6	251	927	1,178	100	94	138
Released	0	0	0	0	7.1	28.3	26.6	19.9	43.6	12.1	28.9	7.2	0	174	174	0	37	37
June																		
Retained	180.2	647.0	13.2	0.2	21.9	103.5	182.3	13.1	601.8	271.6	47.0	38.0	840	1,279	2,120	88	111	142
Released	8.1	38.0	0	0.2	0.7	5.2	0	0.9	55.0	37.9	6.3	1.6	46	108	154	19	21	28
July																		
Retained	153.7	1110.6	25.7	0.2	24.1	114.1	109.7	2.0	542.2	81.3	35.9	112.8	1,290	1,022	2,312	89	89	126
Released	7.6	49.3	0	0.2	0.8	0	5.4	0	42.5	26.7	2.9	0	57	79	136	28	19	34
August																		
Retained	214.0	527.1	218.0	0	11.4	77.9	82.3	28.7	318.7	179.7	17.1	57.9	959	774	1,733	112	98	148
Released	22.0	147.1	0	0	22.8	61.2	0	1.7	12.6	0	3.9	0	169	102	271	76	32	82
TOTAL																		
Retained	638	2,446	257	1	72	387	448	109	1,902	677	164	242	3,341	4,003	7,343	196	197	277
Released	38	234	0	0	31	95	32	22	154	77	42	9	272	462	734	83	57	101
Catch	676	2,680	257	2	103	482	480	132	2,056	754	206	251	3,613	4,465	8,078	213	205	295

Table 22. Total (lodge + non-lodge) estimated catch of other groundfish, by month and subarea, shown separately for released and retained fish, 2021. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	22.5	40.3	0	0	10.7	24.8	6.7	0	3.6	12.1	11.6	4.0	63	73	136	23	16	28
Released	0	0	0	0	7.1	24.8	0	11.4	50.8	0	1.9	0.7	0	97	97	0	20	20
June																		
Retained	9.7	70.2	0	0.2	7.1	45.4	5.1	0.9	78.5	28.9	33.2	5.2	80	205	285	34	29	45
Released	4.0	5.3	0	2.0	20.3	190.9	9.1	1.0	56.3	3.7	8.1	0.2	9	291	301	5	108	109
July																		
Retained	24.9	71.0	38.2	0.2	24.1	89.4	24.2	11.2	157.3	28.6	32.0	31.9	134	399	533	35	51	61
Released	19.6	17.4	30.6	2.7	90.1	108.5	12.1	1.0	227.9	13.4	12.6	4.4	68	473	540	25	85	88
August																		
Retained	58.6	33.2	23.5	0	24.7	66.7	94.6	15.2	134.2	0	44.7	0	115	380	495	27	49	56
Released	36.6	4.7	102.0	2.7	119.5	344.8	61.7	6.4	33.5	263.5	42.1	0	143	874	1,018	42	132	139
TOTAL																		
Retained	116	215	62	0	67	226	131	27	374	70	122	41	392	1,057	1,449	60	78	99
Released	60	27	133	7	237	669	83	20	369	281	65	5	220	1,735	1,955	49	192	198
Catch	176	242	194	8	304	895	213	47	742	350	186	46	612	2,792	3,405	78	207	221

Table 23. Total (lodge + non-lodge) estimated rockfish catch, by month and subarea, shown separately for released and retained fish, 2021. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	5.6	10.1	11.8	7.8	74.9	123.9	75.5	22.7	301.2	58.0	79.1	23.5	28	767	794	14	73	74
Released	16.9	30.3	0	13.0	278.3	180.6	108.7	17.0	635.2	60.4	110.0	29.6	47	1,433	1,480	19	146	148
June																		
Retained	142.2	542.4	64.5	0.2	62.0	253.8	141.8	9.0	544.7	122.0	37.9	15.3	749	1,187	1,936	92	90	129
Released	237.1	326.2	43.6	0	63.7	230.6	174.9	17.0	832.6	166.0	46.5	9.1	607	1,540	2,147	133	138	192
July																		
Retained	247.8	1466.5	51.0	1.4	95.7	299.9	240.8	4.1	923.9	84.9	50.9	114.7	1,765	1,816	3,582	122	112	166
Released	158.6	339.8	21.9	0.3	118.6	436.2	315.4	1.0	1018.6	101.5	56.2	53.7	520	2,102	2,622	84	186	204
August																		
Retained	84.2	317.9	70.6	0	55.0	189.1	300.4	55.8	540.9	215.6	17.1	47.4	473	1,421	1,894	61	120	135
Released	212.3	180.3	329.7	0	70.2	394.8	283.9	76.8	826.0	179.7	25.0	5.3	722	1,862	2,584	110	217	243
TOTAL																		
Retained	480	2,337	198	9	288	867	758	92	2,311	480	185	201	3,015	5,191	8,206	165	201	260
Released	625	877	395	13	531	1,242	883	112	3,312	508	238	98	1,897	6,937	8,833	193	349	399
Catch	1,105	3,213	593	23	819	2,109	1,641	204	5,623	988	423	299	4,911	12,127	17,039	254	403	476

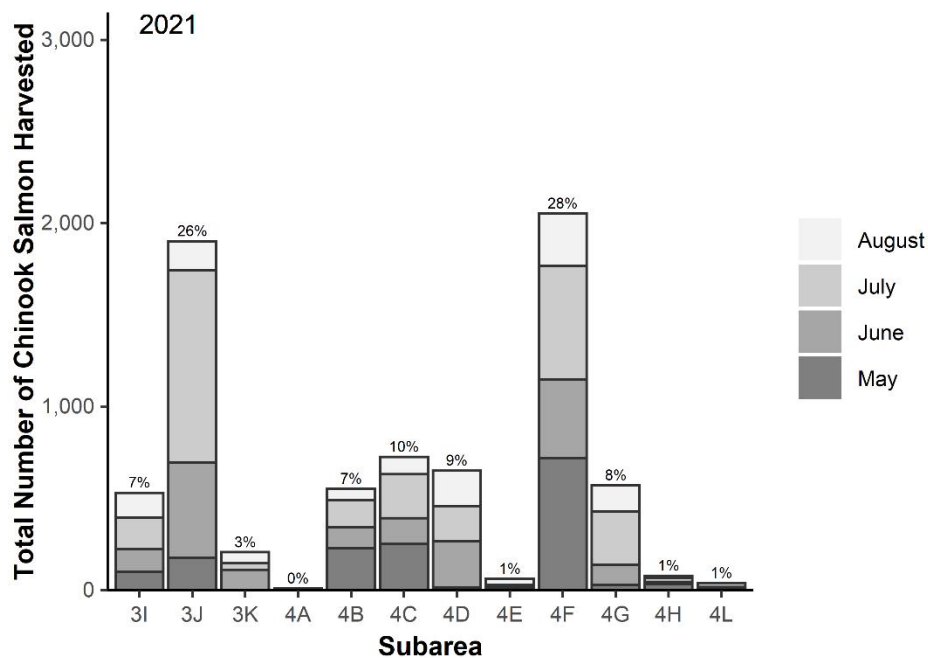


Figure 9. Total (lodge + non-lodge) estimated Chinook Salmon harvest, by month and subarea, 2021.

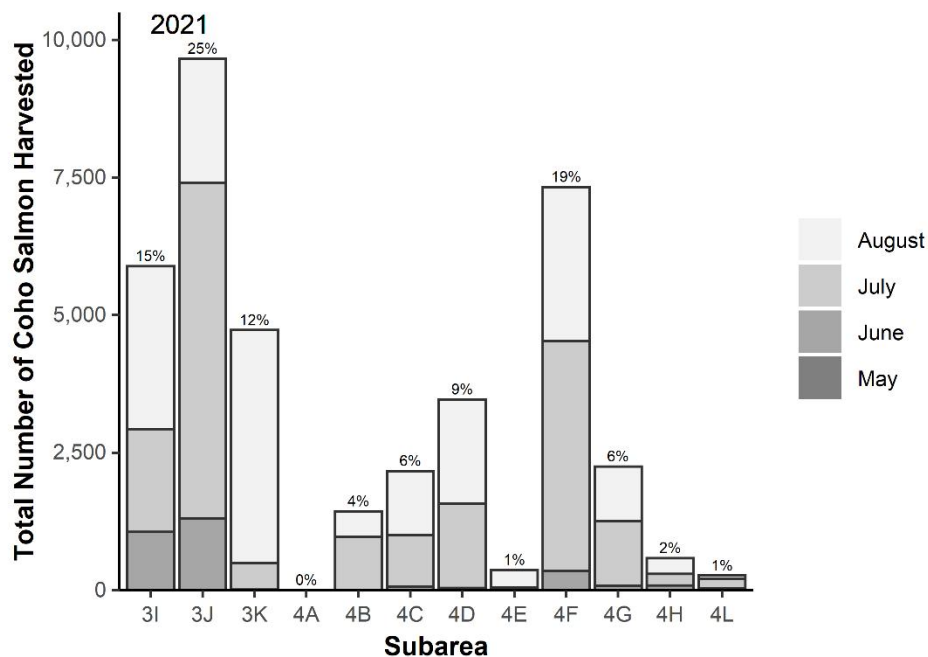


Figure 10. Total (lodge + non-lodge) estimated Coho Salmon harvest, by month and subarea, 2021.

Table 24. Kruskal Wallis tests for differences in harvest (fish caught per boat trip per subarea) among months, by species, 2021. Subareas treated as replicates.

Species	Kruskal-Wallis Statistics			Median Monthly Harvest			
	χ^2	df	P	May	June	July	August
Chinook Salmon	1.9	3	0.59	30.9	105.8	149.1	76.7
Chum Salmon	4.3	3	0.23	0	0	0	0
Coho Salmon	29.9	3	< 0.0001	0	55.0	951.3	1081.0
Pink Salmon	22.6	3	0.0001	0	2.0	54.6	36.0
Sockeye Salmon	6.5	3	0.09	0	0	0	0
Arrow Flounder	6.0	3	0.11	0	0	0	0
Big Skate	2.0	3	0.56	0	0	0	0
Cabezon	3.7	3	0.30	0	0	0	0
English Sole	0.0	3	1	0	0	0	0
Greenling	7.2	3	0.07	0	0	1.6	0
Lingcod	0.4	3	0.94	69.3	75.3	91.8	80.1
Pacific Cod	7.3	3	0.06	3.6	8.4	22.3	26.6
Pacific Halibut	7.7	3	0.054	54.1	208.8	327.3	301.7
Ratfish	0.0	3	1	0	0	0	0
Rock Sole	9.4	3	0.025	0	0	0	0
Sablefish	3.4	3	0.33	0	0	0	0
Starry Flounder	2.3	3	0.52	0	0	0	0
Black Rockfish	4.1	3	0.25	20.9	47.7	45.8	31.3
Boccacio Rockfish	2.1	3	0.54	0	0	0	0
Canary Rockfish	3.0	3	0.39	0	0	0	0
China Rockfish	4.6	3	0.20	0	1.4	2.0	0
Copper Rockfish	3.2	3	0.37	0	0.7	0.2	0.7
Dusky Rockfish	3.5	3	0.32	0	0	0	0
Quillback Rockfish	5.0	3	0.17	11.0	31.4	45.4	36.3
Redband Rockfish	3.0	3	0.39	0	0	0	0
Redstripe Rockfish	2.0	3	0.56	0	0	0	0
Tiger Rockfish	2.8	3	0.43	0	0.3	2.4	0
Widow Rockfish	2.8	3	0.43	0	0	0	0
Yelloweye Rockfish	4.1	3	0.25	0	0	0	0
Yellowtail Rockfish	1.0	3	0.81	0	0.06	0.5	0.8

An estimated 16,671 (SE = 359) Pacific Halibut were harvested in Areas 3 or 4 during the 2021 study period (Table 20; Figure 11). Most of the harvest could be attributed to effort in 3J (28%) and 4F (22%), and in subareas 3I, 4C, and 4G (together another 30%). Harvest varied among months (lowest in May, highest in July), but the effect was not statistically significant ($\chi^2 = 7.7$, df = 3, $P = 0.054$; Table 24). We estimated that 8,116 (SE = 528) Pacific Halibut were released (33% of the total catch).

From May to August 2021, an estimated 7,343 (SE = 277) Lingcod were harvested, with relatively low harvest in May (16%) as compared to the other months (24–31%), though the differences were not statistically significant. ($\chi^2 = 0.4$, df = 3, $P = 0.94$; Table 21; Figure 12). The harvest occurred largely in subarea 3J (33%) and 4F (26%), followed by subareas 3I and 4G (9% each). Of the captured Lingcod, 9% were released (734 fish, SE = 101).

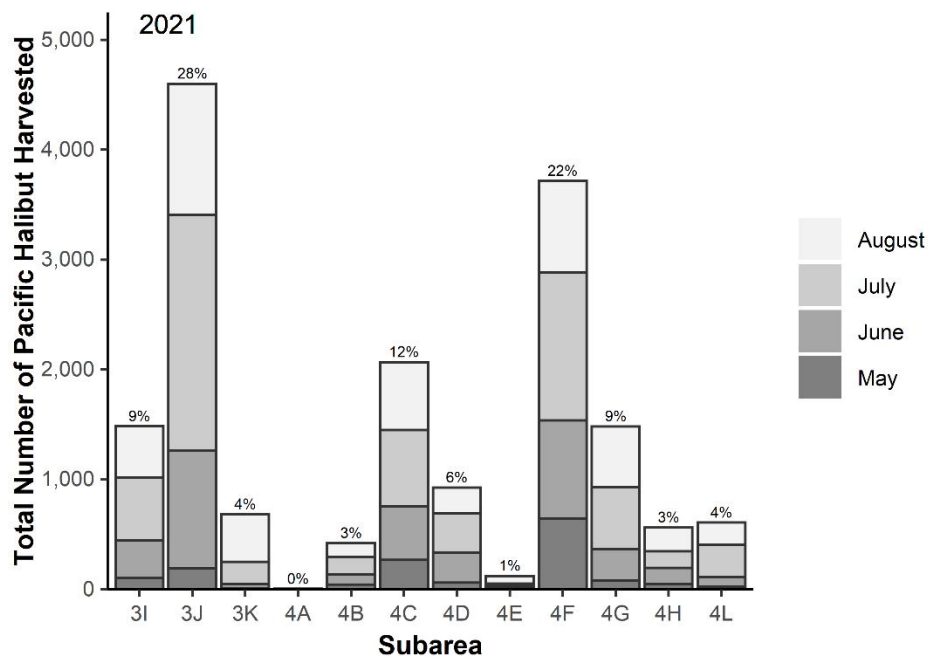


Figure 11. Total (lodge + non-lodge) estimated Pacific Halibut harvest, by month and subarea, 2021.

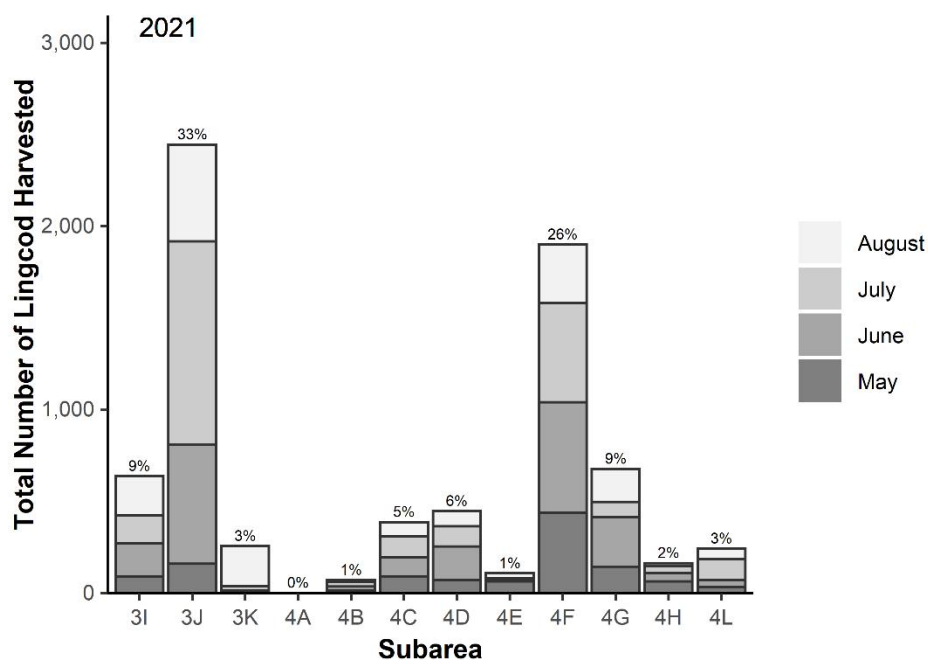


Figure 12. Total (lodge + non-lodge) estimated Lingcod harvest, by month and subarea, 2021.

An estimated 1,449 (SE = 99) other ground fish were harvested from May to August 2021, of which 79% (i.e., 1,150 fish, SE = 88) were Pacific Cod. The remaining harvest included Sablefish (*Anoplopoma fimbria*; 9%), greenlings (*Hexagrammidae*; 5%), Cabezon (*Scorpaenichthys marmoratus*, 3%), and a variety of other species (< 20 individuals retained per species). Over the study period, it is estimated that 542 (SE = 121) Pacific Cod were released (24% of the total catch of Pacific Cod).

Rockfish harvest was estimated at 8,206 fish (SE = 260; Table 23), composed mostly of Black (44%), Quillback (32%), unknown (13%) and eleven other rockfish species. In all, 8,833 (SE = 399) were released. Of the 778 Yelloweye Rockfish caught, all but 11 (99%) were released.

Biosampling

During interviews, observers were told about 1,820 retained Chinook Salmon, and were permitted to handle 614 fish. Too few fish were sexed ($n=12$) to provide a robust estimate of sex ratio (Table 25). The median fork length of biosampled Chinook Salmon ($n = 478$) was 71 cm (95% of observations were between 52 and 92 cm; Figure 13). In all, 74 adipose-clipped individuals were encountered (Table 25), including 40 from standardized sampling methods (complete samples collected from anglers who were asked if their catch could be biosampled before any other interactions took place; Appendix 5) and 34 from non-standard interactions (“non-standard interactions” were interviews for which anything other than complete access was given to all harvested fish). Of the 74, 65% ($n=48$) had their heads collected and labelled for submission to the Salmon Head Recovery Program (Table 25). Of the standardized samples ($n=408$), 9.8% were adipose-clipped (Appendix 5). Scale samples were obtained from 72% ($n=443$) of the biosampled Chinook Salmon (Table 25).

For Coho Salmon, observers were told about 8,645 retained fish, and were permitted to handle 1,164 fish. Too few fish were sexed ($n=5$) to provide a robust estimate of sex ratio (Table 25). The median Coho Salmon fork length ($n = 1,160$) was 62 cm (95% of observations were between 50 and 71 cm; Figure 13). In all, 13 adipose-clipped individuals were encountered (Table 25), only one from standardized sampling interviews. Thirteen heads were collected for submission to the Salmon Head Recovery Program (Table 25). Of the standardized samples ($n=498$), one (0.2%) was adipose-clipped (Appendix 5). Tissue samples were obtained from 99% of the biosampled Coho Salmon (Table 25).

Of the 4,149 Pacific Halibut that anglers reported as retained, 1,073 were handled (Table 25). All were measured, but none were sexed. The median size was 79 cm (95% of observations were between 60 and 121 cm; Figure 13). Average net and round weights, calculated from average lengths using regression formulas, were 5.8 kg (12.8 lb) and 7.7 kg (16.9 lb) in Area 3, and 5.4 kg (11.9 lb) and 7.2 kg (15.8 lb) in Area 4, respectively (Table 26).

Table 25. Biosampling results, 2021.

	Chinook Salmon		Coho Salmon		Pacific Halibut	
	Number	Percent	Number	Percent	Number	Percent
<i>Number Examined</i>	614		1164		1073	
<i>Subarea</i>						
3I	22	4%	119	10%	25	2%
3J	54	9%	275	24%	139	13%
3K	5	1%	48	4%	8	1%
4A	9	1%			2	0%
4B	90	15%	92	8%	39	4%
4C	79	13%	66	6%	191	18%
4D	88	14%	169	15%	97	9%
4E	1	0%	3	0%	4	0%
4F	213	35%	264	23%	330	31%
4G	21	3%	40	3%	70	7%
4H	32	5%	88	8%	163	15%
4L					5	0%
<i>Sex</i>						
Male	6	50%	1	20%		
Female	6	50%	4	80%		
Undeterminable			1			
Unknown	602		1158		1073	
<i>Trip Type</i>						
Guided	214	35%	484	42%	301	28%
Unguided	400	65%	680	58%	772	72%
<i>Adipose Clips</i>						
Clipped	74	12%	13	1%		
Unmarked	538	88%	1120	99%		
Unknown						
<i>Head Tagged</i>						
Yes	48	8%	13	1%		
No	566	92%	1151	99%		
<i>Scale Samples</i>			<i>Whatman Samples</i>			
Yes	443	72%	1152	99%		
No	171	28%	12	1%		
<i>Flesh Colour</i>						
Red	136	89%				
White	13	9%				
Marble	3	2%				
Unknown						

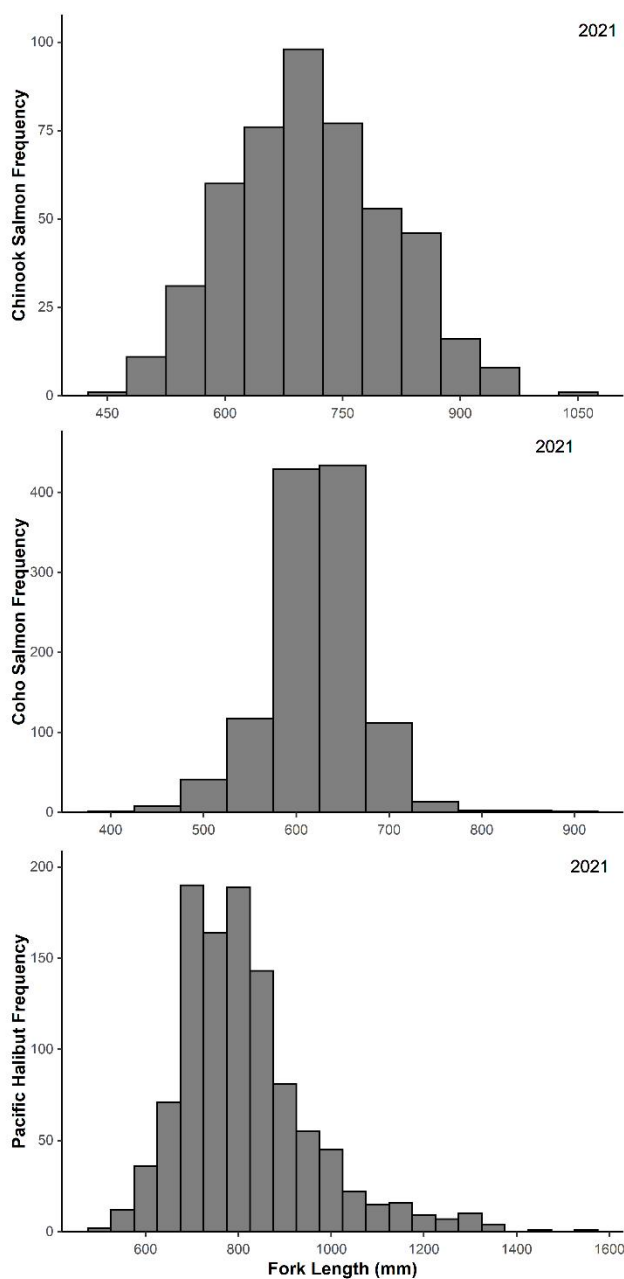


Figure 13. Length-frequency distributions of biosampled Chinook ($n = 478$) and Coho ($n = 1160$) salmon, and Pacific Halibut ($n = 1073$), 2021. Note that X and Y axis scales vary between plots.

Table 26. Average weight, and harvest (numbers and biomass) of Pacific Halibut in Area 3 and 4, 2021.

	Stat Area 3				Stat Area 4			
	Net Weight		Round Weight		Net Weight		Round Weight	
	kg	lb	kg	lb	kg	lb	kg	lb
<i>Creel Interview (Non-Lodge) Average Weights</i>								
<i>Guided</i>	5.5	12.1	7.3	16.0	5.9	13.1	7.8	17.3
<i>Unguided</i>	6.3	13.9	8.3	18.4	5.3	11.6	7.0	15.4
<i>Overall</i>	5.8	12.8	7.7	16.9	5.4	11.9	7.2	15.8
<i>Estimated Non-lodge Harvest (number of fish)</i>								
<i>Overall</i>		5,943				9,905		
<i>Estimated Lodge Harvest (number of fish)</i>								
<i>Overall</i>		823				0		
<i>Biomass Harvested</i>								
<i>Non-lodge</i>	34,467	75,987	45,648	100,636	53,688	118,361	71,103	156,754
<i>Lodge</i>	4,773	10,523	6,321	13,936	0	0	0	0
<i>Overall</i>	39,240	86,510	51,969	114,572	53,688	118,361	71,103	156,754

Harvested Halibut Biomass

Based on the average weights of biosampled Pacific Halibut, and the estimated harvest, it was estimated that total harvested Pacific Halibut net biomass was 92.9 tonnes (204,871 lb), including 39.2 t (86,510 lb) in Area 3 and 53.7 t (118,361 lb) in Area 4. In terms of round weight, Pacific Halibut harvest was 123.1 t (271,326 lb) including 52.0 t (114,572 lb) in Area 3 and 71.1 t (156,754 lb) in Area 4 (Table 26).

DISCUSSION

Creel Estimates

In general, recreational fishing effort and salmon catch in Areas 3 & 4 have been increasing over the last 27 years (Tables 27 and 28). Using published creel data collected since 1995, we found 13 years (1995-96, 2001-02, 2008-09², and 2015-2021) for which effort data were collected in June, July and August (Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017, Robichaud and Addison 2018, 2019, 2020, 2021), and we have therefore limited this meta analyses to those three months. Meta-analysis revealed (Figure 14) that fishing effort has increased steadily (by 216 trips per year on average) since 1995. Effort in 2021 was lower than what would have been predicted from the long-term trend, possibly attributable to the international and domestic travel restrictions associated with the COVID-19 pandemic.

² Creel data from 2010-2014 were collected and analyzed by DFO, and preliminary results were presented as part of their post-season review process. We based our meta-analysis on published accounts (written accounts showing methods, results, and discussing the caveats of the results), and hence have not included data from these years.

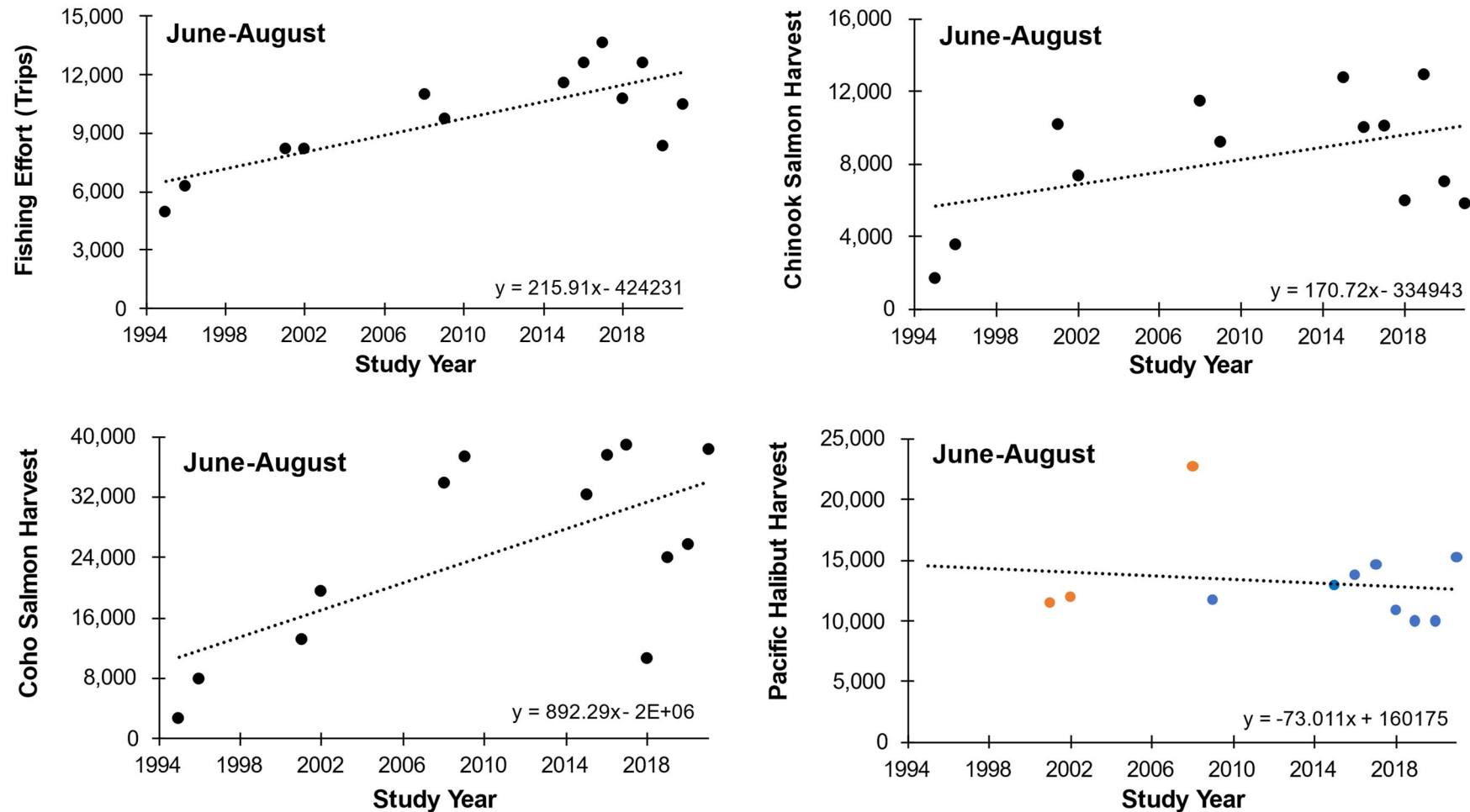


Figure 14. Fishing effort, and harvest of Chinook and Coho salmon, and Pacific Halibut from 1995 to 2021. Annual estimates only include data from June, July and August. Only years with data from all three months are included. Linear regression equations are shown in the lower right of each panel. For Pacific Halibut, bag limits changed after 2008 from two fish per day (orange dots) to one fish per day (blue dots).

Table 27. Monthly recreational fishing effort (boat trips) in Area 3 and 4 from 1995 to 2021, as estimated using creel survey methods.

Month	Year															
	1995	1996	1998	2000	2001	2002	2006	2008	2009	2015	2016	2017	2018	2019	2020	2021
May	965	391	--	--	449	960	--	983	--	--	--	--	--	1,553	1,320	1,785
June	1,590	2,490	--	--	3,801	3,463	--	4,696	3,043	3,424	3,961	4,574	2,873	3,790	1,970	2,507
July	1,394	2,385	1,307	1,842	2,588	2,892	--	3,750	3,804	4,279	5,295	6,044	4,593	4,644	3,745	4,318
Aug	1,927	1,400	1,351	1,327	1,808	1,843	2,593	2,529	2,865	3,881	3,325	2,980	3,304	4,175	2,631	3,642
Sept	864	488	504	954	540	676	758	394	817	--	--	--	--	--	--	--
TOTAL	6,740	7,154	3,162	4,123	9,186	9,834	3,351	12,352	10,529	11,585	12,581	13,598	10,770	14,163	9,665	12,252

Table 28. Monthly Chinook Salmon, Coho Salmon and Pacific Halibut harvest in the Area 3 and 4 recreational fishery from 1995 to 2021, as estimated using creel survey methods.

Month	Year															
	1995	1996	1998	2000	2001	2002	2006	2008	2009	2015	2016	2017	2018	2019	2020	2021
Chinook Salmon Harvest																
May	141	45	--	--	509	522	--	474	--	--	--	--	--	2,234	1,256	1,580
June	907	2,165	--	--	6,630	4,028	--	7,708	4,797	7,071	5,678	5,229	1,273	5,675	2,574	1,834
July	505	1,097	254	1,611	3,260	2,822	--	3,080	3,873	4,016	3,474	3,800	2,964	4,806	2,856	2,797
Aug	245	257	35	108	299	474	1,207	688	503	1,669	891	1,078	1,593	2,437	1,560	1,172
Sept	15	62	6	93	61	59	203	20	4	--	--	--	--	--	--	--
TOTAL	1,813	3,626	295	1,812	10,759	7,905	1,410	11,970	9,177	12,756	10,043	10,108	5,997	15,152	8,247	7,383
Coho Salmon Harvest																
May	0	0	--	--	0	0	--	0	--	--	--	--	--	29	0	0
June	335	474	--	--	1,771	1,332	--	3,913	3,686	4,436	4,389	1,839	126	1,475	1,006	3,038
July	402	3,565	1,937	568	4,736	9,798	--	15,982	17,043	15,101	21,761	22,871	5,758	10,341	16,304	17,685
Aug	1,907	3,771	1,500	2,125	6,488	8,292	15,647	13,791	16,459	12,612	11,200	14,004	4,580	11,974	8,288	17,403
Sept	1,115	1,417	79	790	1,006	1,010	3,215	733	3,136	--	--	--	--	--	--	--
TOTAL	3,759	9,227	3,516	3,483	14,001	20,432	18,862	34,419	40,324	32,148	37,350	38,713	10,464	23,820	25,599	38,126
Pacific Halibut Harvest																
May	--	--	--	--	281	467	--	1,435	--	--	--	--	--	941	867	1,499
June	--	--	--	--	4,872	3,912	--	8,367	3,029	3,551	4,032	4,632	3,407	2,698	2,344	3,725
July	--	--	--	3,963	3,535	4,716	--	8,507	4,466	4,096	6,069	6,563	4,263	3,894	4,150	6,518
Aug	--	--	1,801	2,530	3,085	3,247	8,178	5,865	4,142	5,201	3,670	3,410	3,183	3,435	3,504	4,929
Sept	--	--	328	1,224	222	615	1,812	774	1,205	--	--	--	--	--	--	--
TOTAL	--	--	2,129	7,717	11,995	12,957	9,990	24,948	12,842	12,848	13,771	14,606	10,853	10,967	10,865	16,671

Over the same period, Chinook and Coho salmon harvests have increased by 171 and 892 fish per year, on average (Table 28, Figure 14). Chinook Salmon harvest in 2021 was low relative to what expected given the observed level of effort, probably resulting from the management measures which imposed a non-retention period from 19 June to 11 July 2021. The 2021 Coho Salmon harvest was among the highest observed over the 27-year period (values were 1.5 times higher than in 2020, and > 14 times higher than in 1995). Harvest was higher than that expected given the long-term trend, which is more concerning given the lower-than-expected effort observed in 2021.

The Lingcod harvest (June to August) from 2018 to 2021 (4,270 in 2018, 4,295 in 2019, 4,064 in 2020, and 6,165 in 2021) was higher than in other recent years (3,052 in 2015, 1,803 in 2016, and 2,772 in 2017). This, despite a coast-wide reduction in the Lingcod bag limit from three to two fish starting in May 2018. While harvest levels appeared to have been relatively stable since 2018 (despite a general trend toward increasing effort), the elevated harvest value from 2021 could potentially cause concern.

Pacific Halibut harvest did not show a strong among-year trend (Figure 14). While fishing effort increased over the 27-year period, the total harvest of Pacific Halibut in recent years (2009, 2015-2021) has been very similar to that from 2001-2002; Table 28). This is very likely the result of effective management measures that were enacted between 2008 and 2009 (including the reduction in the daily bag limit from two to one), that appeared to have reduced the catch of halibut by nearly 50% (Figure 14).

The meta-analysis results make two important points: 1) the effort and CPUE of certain species has increased tremendously over the past ~25 years (even despite the 2020-21 pandemic) potentially endangering stocks if left unchecked; and 2) management decisions from the recent past appear to have been effective, indicating that effective tools are available to keep effort and CPUE increases from going unchecked.

May Effort

In our report for the 2018 study year (Robichaud and Addison 2019), we made the case that substantial angling activity was being missed by starting the creel survey in June each year. Surveys in 1995 and 1996 (Van Tongeren and Winther 2010) included May, which accounted for 14% and 5% of the May-August effort, and for 8% and 1% of the May-August Chinook harvest. May surveys occurred again in 2001, 2002, and 2008, representing 5%, 10%, and 8% of the May-August effort, respectively (and 5%, 7%, and 4% of the May-August Chinook harvest). Beginning in 2019, we did start the survey in May, and found that May effort (1,553 boat trips in 2019, 1,320 in 2020, and 1,785 in 2021) accounted for 11-15% of the May-August total, and for 15-21% of the May-August total Chinook harvest.

Much of the early season angling targets Chinook Salmon, and occurs during a period in which the Area 3 and 4 stock composition is unknown or poorly resolved. As such, the Department of Fisheries and Oceans Canada (DFO) expressed interest in a May creel survey because genetic samples collected from Chinook Salmon would help them resolve stock composition and help determine exploitation rates on early-run-timing Chinook Salmon stocks. In 2019, during the month of May, 269 Chinook Salmon were biosampled, including 230 scale samples, and 22

tagged heads. In 2020, 144 Chinook Salmon scale samples were collected in May, more than in the following three months combined. In 2021, 145 Chinook Salmon scale samples were collected in May, which represented 33% of the annual total, and submitted 15 tagged heads to the Salmon Head Recovery Program.

Effect of COVID-19

Government-imposed travel and group-size restrictions associated with the COVID-19 pandemic may have been responsible for some of the patterns observed in 2021, though the impacts were less pronounced than in 2020. Fishing effort in 2021 rebounded somewhat from the 20-year low that was observed in 2020, but was still below that which would be expected given the long-term trend for increasing effort levels. Nevertheless, the proportion of trips that were guided in 2021 had rebounded to 24% (similar to previous years; e.g., 24-34% in 2015-19, see Robichaud et al. 2016, 2017, Robichaud and Addison 2018, 2019, 2020) from the lower level observed in 2020 (17% of trips; Robichaud and Addison 2021). In addition, the proportional compliance with biosampling requests in 2021 showed evidence of rebounding to pre-pandemic levels (Figure 15; since the decision by anglers to allow or disallow biosampling is assumed to be random with respect to their general participation in the fishery, there is no reason to expect the biosampling data from 2021 to be biased). Despite restrictions, the observers were able to maintain a normal workload (number of shifts) by working alone, remaining outdoors, and using appropriate PPE (gloves, facial coverings, etc.). Indeed, observers sampled 23% of the estimated number of non-lodge boat trips in 2021, on par with the sampling rates from prior years (e.g., 16-26% in 2015-19, see Robichaud et al. 2016, 2017, Robichaud and Addison 2018, 2019, 2020, 2021).

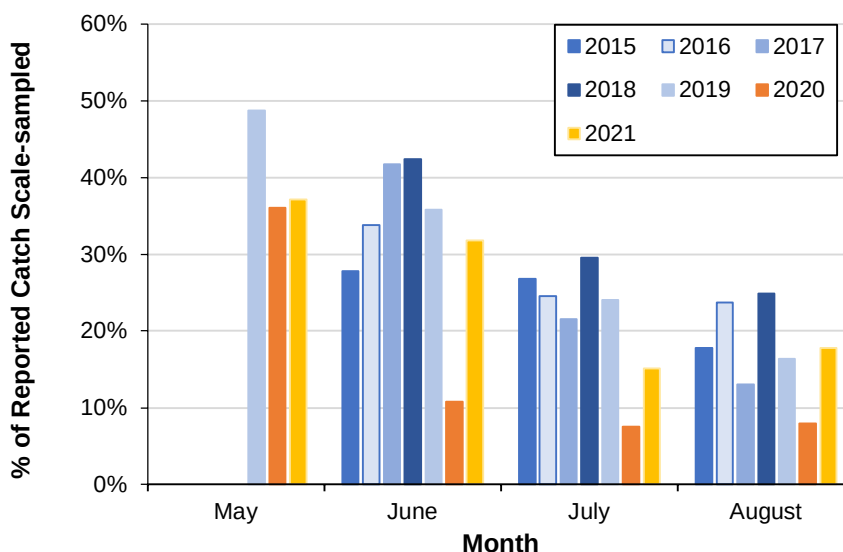


Figure 15. Biosampling compliance (percent of reported Chinook Salmon catch that was sampled for scales) by year.

Precision of the Results

Overall, our level of survey effort, both for CPE and Effort estimation, were adequate, as evidenced by the relatively good precision of the harvest estimates for most of the taxa surveyed. In 2021 our precision goal was to estimate total harvests within 15% of the true value 19 times out of 20, for the main taxa (Chinook Salmon, Coho Salmon and Pacific Halibut), for which harvest estimates were precise to within 3-6%. Pink Salmon, Lingcod, rockfish, and other groundfish estimates were precise to within 13%, 7%, 6%, and 13%, respectively. Chum and Sockeye salmon estimates were not precise to within 15, though the survey was not designed to meet any level of precision for these last species.

The precision of our harvest estimates depended largely on the variability in the CPE and effort data. Differing levels of angler expertise, and day-to-day changes in abundance of the target species, fishing effort, weather, or fishing conditions mean that fishing trips have a wide range of outcomes. Similarly, day-to-day differences in weather or other factors (such as leisure time, day of week, the price of gasoline, etc.) affect the number of anglers that decide to go out on the water on any given day, resulting in variability within the boat-count dataset. Sampling error is another main source of estimation imprecision; and by meeting our precision goals we have demonstrated that we are sampling an adequate proportion of the population.

Accuracy of the Results

The accuracy of the results depends on the quality of data provided by the anglers to the interviewers. Each time an angler is relied upon for unconfirmed data, there is opportunity for results to become inaccurate. For example, in this study, 614 of the 1,820 Chinook Salmon (34%) and 1,164 of the 8,645 Coho Salmon (13%) reported as kept during angler interviews were recorded as observed by the interviewer. While not every fish can be observed³, each one that goes unconfirmed is nevertheless a possible source of inaccuracy in the reported harvest totals.

In this study, the angler effort estimates depend on the accuracy of the aerial survey boat counts and the activity patterns for each month. For example, we have no estimate of the proportion of angling boats that are missed during overflights, where missed boats would contribute to the underestimation of recreational fishing effort. Also, we do not know if all fishing areas are adequately covered by the current flightpath – perhaps there are fishing locations that are hidden behind islands (e.g., Dunira Island or the Moffat Islands), too far from the flightpath for the surveyors to see active fishing boats. On the other hand, our estimate of fishing effort may be overestimated due to our inability to survey and count boats on foggy or bad weather days. We know that during bad weather, fewer anglers go out on the water; yet these same conditions can cause flights to be cancelled, and thus the lowest effort values are sometimes not sampled.

³ Fish may not be observed when anglers refuse to provide access to their fish storage containers for various reasons. Reluctance to provide access may be due to fish already being processed and packed in ice, boat launch traffic congestion causing tension, or COVID-19 concerns, among other reasons.

Another potential source of inaccuracy is the removal of lodge boats from the overflight counts. There is uncertainty around whether all of the active lodge boats were observed by the overflights (e.g., vessels outside of the survey area). There is also uncertainty around the proportion of lodge vessels that were active during the overflights. These factors may influence the accuracy of the estimates of non-lodge fishing effort.

RECOMMENDATIONS

- The catch and effort estimates for the Area 3 and 4 recreational creel surveys for 2010-2014 should be compiled and included in summary tables similar to Table 27 and 28 in this report. If sufficient funding can be obtained, the results from 2010-14 surveys should be compiled into a manuscript report. This additional information will help provide context for the current results, and help clarify how rates of effort and harvest have changed in recent years.
- We recommend that future surveys continue to run for four months (May to August) rather than three (June to August). May surveys will help document and quantify angling effort and harvest during that month. The Department of Fisheries and Oceans Canada (DFO) has expressed interest in a May creel survey because genetic samples collected from Chinook Salmon will help resolve the currently unknown early-season Area 3 and 4 stock composition, which will help determine exploitation rates on early-run-timing Chinook Salmon stocks.
- Trailer counts should be conducted in April 2022 for comparison with April trailer counts in 2019-2020 and continued throughout the creel survey season as in recent years.

ACKNOWLEDGEMENTS

We would like to thank the creel surveyors (Anke De Wit, Alaina Pyde, Brent Merriam, Chris Fraser, Lisa Grant and Brandie Loranger) for collecting large amounts of data and bio-samples from anglers, despite COVID-related risks. Thanks to Pacific Salmon Commission, who provided support for this work through the Northern Fund. Thanks to Darren Chow and Ryan Whitmore for acting as liaison between our team and DFO. Thanks also go to the administrative staff at LGL and NCSFNSS, whose work made it possible for this study to be conducted. We also would like to thank Karl English for providing useful comments and edits on previous drafts of this report. Last but not least, we thank the many anglers, charter operators, and fishing lodges who participated in the survey.

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APPENDICES

APPENDIX 1: DATA FORMS

2021 AREA 3/4 CREEL SURVEY
AERIAL SURVEY VESSEL ENUMERATION FORM

DATE: _____

FLIGHT#: _____

SURVEYOR: _____

Note: only survey shaded subareas when conditions permit

Sub Area	RIGHT FRONT COUNT				TOTAL COUNT				FLIGHT TIME		SEA STATE (1-4)
	SPORT (Fishing)	SPORT (Moving)	COMM. (Fishing)	COMM. (Moving)	S (F)	S (M)	C (F)	C (M)	START	FINISH	
A											
C											
B											
C											
D											
E											
C											
F											
E											
F											
G											
L											
G											
J											
I											
K											
I											
H											
C											
A											

WEATHER / SEA STATE / COMMENTS: _____

SIGNATURE: _____

SHIFT SUMMARY SHEET

Landing Site: _____ Interviewer Name: _____

Interviewer Code: _____ Date: _____

Shift Code: _____ Shift Times: _____

Time Blocks	
From - To	TB
Before 7:00	1
7:00 - 7:59	2
8:00 - 8:59	3
9:00 - 9:59	4
10:00 - 10:59	5
11:00 - 11:59	6
12:00 - 12:59	7
13:00 - 13:59	8
14:00 - 14:59	9
15:00 - 15:59	10
16:00 - 16:59	11
17:00 - 17:59	12
18:00 - 18:59	13
19:00 - 19:59	14
20:00 - 20:59	15
21:00 - 21:59	16

* Fill in using numbers only.

Time Block	Fishing Conditions	Boats Missed	Not Fishing	Fishing

Totals:

--	--	--

Comments: _____

Trailer Count: _____ Time: _____

Database Tally # :

(office use only)

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APPENDIX 2: HPE ESTIMATES (AVERAGE NUMBER OF FISH HARVESTED PER BOAT TRIP) BY PERIOD, TRIP TYPE, SUBAREA AND SPECIES.

Average number of salmon harvested per boat trip in May 2021, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	n/a	n/a	n/a		0.46	1.91	2.08	2.08	2.14	0.67	0.88	0.79
Unguided	0.82	0.82		0.09	0.70	0.64	0.25	0.43	1.41	0.44	0.45	0.44
Combined	0.71	0.71		0.07	0.71	0.80	0.24	0.43	1.62	0.50	0.57	0.53
Chum Salmon												
Guided	n/a	n/a	n/a									
Unguided												
Combined												
Coho Salmon												
Guided	n/a	n/a	n/a									
Unguided												
Combined												
Pink Salmon												
Guided	n/a	n/a	n/a									
Unguided												
Combined												
Sockeye Salmon												
Guided	n/a	n/a	n/a									
Unguided												
Combined												
Unknown Salmon												
Guided	n/a	n/a	n/a									
Unguided					0.01							
Combined					0.01							

Average number of salmon harvested per boat, 1-18 June 2021, by trip type and subarea. Zeros hidden.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	2.50	2.82	2.50		0.37	1.71	2.79	2.18	1.73	1.67	2.00	1.93
Unguided	1.36	1.33	0.83	0.02	1.13	0.50	0.94	1.03	1.09	0.67	0.64	0.65
Combined	1.38	2.29	0.86	0.01	1.21	0.73	1.73	1.44	1.29	1.00	1.05	1.10
Chum Salmon												
Guided												
Unguided												
Combined												
Coho Salmon												
Guided	0.29	0.36	0.29				0.13	0.07	0.03		0.17	0.07
Unguided												
Combined		0.24					0.05	0.03	0.01		0.05	0.03
Pink Salmon												
Guided												
Unguided												
Combined												
Sockeye Salmon												
Guided												
Unguided												
Combined												
Unknown Salmon												
Guided												
Unguided												
Combined												

Average number of salmon harvested per boat trip, 19 June - 11 July 2021, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided												
Unguided								0.02	0.02			
Combined								0.01	0.02			
Chum Salmon												
Guided												
Unguided	0.05										0.03	0.02
Combined	0.03										0.02	
Coho Salmon												
Guided	9.38	6.31	6.97		0.08	1.67	1.43	3.81	4.48	4.00	1.77	2.80
Unguided	3.36	3.48			0.08	0.29	0.50	1.25	1.52	0.52	1.73	1.37
Combined	5.89	5.43	0.29		0.08	0.37	0.69	1.83	2.21	1.42	1.75	1.75
Pink Salmon												
Guided	0.56	0.31	0.37					0.03	0.04	0.25	0.13	
Unguided	0.41	0.04		0.01		0.03		0.10	0.13	0.04	0.05	0.04
Combined	0.47	0.23		0.00		0.03		0.08	0.11	0.10	0.07	
Sockeye Salmon												
Guided												
Unguided												
Combined												
Unknown Salmon												
Guided												
Unguided												
Combined												

Average number of salmon harvested per boat trip, 12-31 July 2021, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	0.83	2.24	2.11		1.22	1.36	0.71	1.24	1.47	2.71	1.38	0.45
Unguided	1.10	1.22	0.22	0.01	0.65	0.36	0.66	0.43	0.62	0.93	0.72	0.77
Combined	1.04	1.85	0.21	0.00	0.75	0.45	0.66	0.63	0.76	1.52	0.88	0.45
Chum Salmon												
Guided												
Unguided												
Combined												
Coho Salmon												
Guided	12.00	9.88	9.99		11.06	3.91	9.88	8.19	7.22	6.86	7.46	2.82
Unguided	3.33	6.17	2.61	0.01	3.46	1.38	4.33	5.14	4.24	5.07	4.38	4.57
Combined	5.26	8.48	2.79	0.00	4.83	1.62	5.21	6.25	4.70	5.67	5.15	2.82
Pink Salmon												
Guided		0.33	0.30		0.11	0.73	0.35	0.26	0.22	0.14	0.08	0.45
Unguided	0.57	0.30	0.56		0.15	0.18	0.24		0.22	0.14	0.15	0.15
Combined	0.44	0.32	0.53		0.14	0.23	0.26		0.22	0.14	0.13	0.45
Sockeye Salmon												
Guided												
Unguided		0.02										
Combined		0.01										
Unknown Salmon												
Guided												
Unguided						0.01						
Combined						0.01						

Average number of salmon harvested per boat trip in August 2021, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	0.39	0.51	0.13		1.29	0.29	0.60	0.77	0.80	0.86	0.33	
Unguided	0.59	0.50	0.14		0.25	0.14	0.53	0.47	0.40	0.33	0.03	0.13
Combined	0.47	0.51	0.13		0.33	0.15	0.53	0.52	0.49	0.48	0.15	
Chum Salmon												
Guided												
Unguided												
Combined												
Coho Salmon												
Guided	10.48	6.94	12.54		9.86	7.00	7.20	7.14	7.13	5.00	4.86	1.83
Unguided	7.74	6.79	5.52		1.95	1.48	5.10	4.57	4.16	2.67	3.39	2.96
Combined	9.47	6.88	8.70		2.54	1.81	5.22	4.96	4.80	3.32	3.96	1.22
Pink Salmon												
Guided	0.39	0.80	0.50				0.40	0.51	0.53		0.19	
Unguided	0.44	0.38	0.41		0.08	0.12	0.14	0.17	0.19		0.21	0.13
Combined	0.41	0.63	0.45		0.07	0.11	0.16	0.23	0.27		0.20	
Sockeye Salmon												
Guided												0.17
Unguided							0.04	0.02				
Combined							0.03	0.01				0.11
Unknown Salmon												
Guided												
Unguided							0.04	0.02				
Combined							0.03	0.01				

Average number of groundfish harvested per boat trip in May 2021, by trip type and subarea. Zeros hidden.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided												
Unguided						0.02	0.01	0.01				
Combined						0.02	0.01	0.00				
Big Skate												
Guided												
Unguided												
Combined												
Cabezon												
Guided								0.06	0.07			
Unguided		0.04						0.01	0.02			
Combined		0.02						0.02	0.03			
Greenling												
Guided	0.02											
Unguided						0.02	0.01	0.01	0.01			
Combined	0.01					0.02	0.01	0.01	0.01			
Lingcod												
Guided	0.74	1.83	0.79					0.63	0.73	1.14		1.67
Unguided	0.22	0.50			0.07	0.13	0.24	0.42	0.50	0.39	0.39	0.39
Combined	0.55	1.29	0.36		0.06	0.12	0.23	0.45	0.55	0.60	0.24	1.22
Pacific Cod												
Guided	0.28	0.03	0.08		0.29			0.23	0.27		0.86	
Unguided	0.07	0.21	0.03		0.13	0.07	0.20	0.16	0.13		0.48	0.30
Combined	0.21	0.10	0.06		0.14	0.07	0.19	0.17	0.16		0.63	
Pacific Halibut												
Guided	1.89	4.97	1.17		2.43	0.29	0.40	1.91	2.17	1.86	3.48	5.00
Unguided	0.81	1.71	0.66	0.02	0.53	1.00	0.66	0.96	1.22	1.83	2.73	2.43
Combined	1.49	3.64	0.89	0.01	0.67	0.96	0.65	1.10	1.42	1.84	3.02	4.22
Rock Sole												
Guided												
Unguided							0.01	0.01	0.01			
Combined							0.01	0.01	0.01			
Pacific Halibut												
Guided	1.89	4.97	1.17		2.43	0.29	0.40	1.91	2.17	1.86	3.48	5.00
Unguided	0.81	1.71	0.66	0.02	0.53	1.00	0.66	0.96	1.22	1.83	2.73	2.43
Combined	1.49	3.64	0.89	0.01	0.67	0.96	0.65	1.10	1.42	1.84	3.02	4.22
Sablefish												
Guided												
Unguided								0.02	0.03			
Combined								0.01	0.02			
Starry Flounder												
Guided												
Unguided												
Combined												
Unknown Groundfish												
Guided												
Unguided							0.04	0.02	0.01			
Combined							0.03	0.02	0.01			

Average number of groundfish harvested per boat trip, 1-18 June 2021, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided												
Unguided								0.01	0.01			
Combined								0.01	0.01			
Big Skate												
Guided												
Unguided						0.03						
Combined						0.03						
Cabezon												
Guided												
Unguided												
Combined												
Greenling												
Guided												
Unguided												
Combined												
Lingcod												
Guided	1.14	1.45	1.14		0.16	0.79	1.58	1.96	2.24	3.50	2.67	3.14
Unguided	1.09	0.17			0.18	0.20	0.53	1.01	1.19	1.33	0.43	0.85
Combined	0.92	1.00			0.23	0.31	0.98	1.35	1.52	2.06	1.10	1.65
Pacific Cod												
Guided	0.29	0.36	0.29					0.04	0.06		0.33	0.14
Unguided	0.09					0.03		0.11	0.16	0.17		0.08
Combined	0.08	0.24				0.03		0.08	0.13	0.11	0.10	0.10
Pacific Halibut												
Guided	2.14	2.64	2.14		0.48	1.86	2.13	1.98	1.88	1.67	1.67	2.14
Unguided	0.91	1.33	0.17		0.56	1.08	0.66	1.37	1.71	2.00	1.64	1.81
Combined	0.85	2.18	0.14		0.86	1.23	1.29	1.59	1.77	1.89	1.65	1.93
Rock Sole												
Guided												
Unguided												
Combined												
Pacific Halibut												
Guided	2.14	2.64	2.14		0.48	1.86	2.13	1.98	1.88	1.67	1.67	2.14
Unguided	0.91	1.33	0.17		0.56	1.08	0.66	1.37	1.71	2.00	1.64	1.81
Combined	0.85	2.18	0.14		0.86	1.23	1.29	1.59	1.77	1.89	1.65	1.93
Sablefish												
Guided												
Unguided						0.07		0.01	0.01			
Combined						0.05		0.01	0.01			
Starry Flounder												
Guided												
Unguided												
Combined												
Unknown Groundfish												
Guided												
Unguided												
Combined												

Average number of groundfish harvested per boat trip, 19 June - 11 July 2021, by trip type and subarea.
Zeros hidden.

		Subarea											
Species / Trip Type		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder													
Guided													
Unguided													
Combined													
Big Skate													
Guided													
Unguided													
Combined													
Cabezon													
Guided													
Unguided			0.04					0.04	0.01				
Combined			0.01					0.03	0.01				
Greenling													
Guided													
Unguided									0.01	0.01		0.02	0.01
Combined									0.01	0.01		0.01	
Lingcod													
Guided	0.75	1.45	1.26		0.01	0.17	2.29	0.78	0.36	1.38	0.71	2.00	
Unguided	0.05	0.83	0.33	0.01		0.23	0.50	0.68	0.74	0.61	0.64	0.63	
Combined	0.34	1.26	0.29	0.00		0.23	0.86	0.70	0.65	0.81	0.66	1.50	
Pacific Cod													
Guided		0.18	0.13		0.02	0.50		0.34	0.44	1.00	1.52	0.40	
Unguided	0.05				0.34	0.08	0.07	0.07	0.07	0.04	0.17	0.13	
Combined	0.03	0.12			0.33	0.11	0.06	0.13	0.16	0.29	0.61	0.25	
Pacific Halibut													
Guided	2.06	3.27	2.94		0.16	2.17	0.71	2.84	3.44	2.75	3.94	5.40	
Unguided	1.09	1.26	0.50	0.02	0.44	1.24	2.14	1.68	1.54	0.96	1.78	1.59	
Combined	1.50	2.65	0.43	0.01	0.52	1.29	1.86	1.94	1.98	1.42	2.48	4.25	
Rock Sole													
Guided													
Unguided													
Combined													
Pacific Halibut													
Guided	2.06	3.27	2.94		0.16	2.17	0.71	2.84	3.44	2.75	3.94	5.40	
Unguided	1.09	1.26	0.50	0.02	0.44	1.24	2.14	1.68	1.54	0.96	1.78	1.59	
Combined	1.50	2.65	0.43	0.01	0.52	1.29	1.86	1.94	1.98	1.42	2.48	4.25	
Sablefish													
Guided									0.03	0.04			
Unguided						0.02			0.02	0.02		0.02	0.01
Combined						0.02			0.02	0.03		0.01	
Starry Flounder													
Guided				0.01	0.01								
Unguided								0.04	0.01				
Combined				0.00				0.03	0.01				
Unknown Groundfish													
Guided													
Unguided													
Combined													

Average number of groundfish harvested per boat trip, 12-31 July 2021, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided												
Unguided												
Combined												
Big Skate												
Guided												
Unguided												
Combined												
Cabezon												
Guided												
Unguided	0.05						0.01					
Combined	0.04						0.01					
Greenling												
Guided		0.01	0.01		0.22			0.04	0.06			
Unguided	0.05	0.02				0.01	0.01					
Combined	0.04	0.02			0.04	0.01	0.01		0.01			
Lingcod												
Guided	1.17	1.07	1.06		0.39	0.27	0.41	0.63	0.75		0.31	1.64
Unguided	0.14	0.89	0.06		0.06	0.11	0.21	0.29	0.51	0.21	0.18	0.19
Combined	0.37	1.00	0.05		0.12	0.13	0.24	0.25	0.54	0.14	0.21	1.64
Pacific Cod												
Guided		0.11	0.10		0.06	0.27		0.50	0.47	0.14	0.15	0.55
Unguided	0.10	0.02	0.06		0.04	0.08	0.06		0.04		0.08	0.06
Combined	0.07	0.07	0.05		0.04	0.09	0.05	1.25	0.11	0.05	0.10	0.55
Pacific Halibut												
Guided	0.67	3.28	3.05		2.39	1.91	1.24	2.26	2.58	3.86	1.85	4.09
Unguided	2.48	2.15	1.22		0.38	0.70	0.90	0.71	1.05	1.79	1.41	1.51
Combined	2.07	2.85	1.16		0.74	0.81	0.95	1.63	1.29	2.48	1.52	4.09
Rock Sole												
Guided												
Unguided												
Combined												
Pacific Halibut												
Guided	0.67	3.28	3.05		2.39	1.91	1.24	2.26	2.58	3.86	1.85	4.09
Unguided	2.48	2.15	1.22		0.38	0.70	0.90	0.71	1.05	1.79	1.41	1.51
Combined	2.07	2.85	1.16		0.74	0.81	0.95	1.63	1.29	2.48	1.52	4.09
Sablefish												
Guided								0.02	0.03			
Unguided			0.22			0.02			0.04		0.03	0.02
Combined			0.21			0.02			0.04		0.02	
Starry Flounder												
Guided												
Unguided												
Combined												
Unknown Groundfish												
Guided												
Unguided												
Combined												

Average number of groundfish harvested per boat trip in August 2021, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided												
Unguided						0.02	0.01	0.01				
Combined						0.02	0.01	0.00				
Big Skate												
Guided												
Unguided												
Combined												
Cabezon												
Guided								0.06	0.07			
Unguided		0.04						0.01	0.02			
Combined		0.02						0.02	0.03			
Greenling												
Guided	0.02											
Unguided						0.02	0.01	0.01	0.01			
Combined	0.01					0.02	0.01	0.01	0.01			
Lingcod												
Guided	0.74	1.83	0.79					0.63	0.73	1.14		1.67
Unguided	0.22	0.50			0.07	0.13	0.24	0.42	0.50	0.39	0.39	0.39
Combined	0.55	1.29	0.36		0.06	0.12	0.23	0.45	0.55	0.60	0.24	1.22
Pacific Cod												
Guided	0.28	0.03	0.08		0.29			0.23	0.27		0.86	
Unguided	0.07	0.21	0.03		0.13	0.07	0.20	0.16	0.13		0.48	0.30
Combined	0.21	0.10	0.06		0.14	0.07	0.19	0.17	0.16		0.63	
Pacific Halibut												
Guided	1.89	4.97	1.17		2.43	0.29	0.40	1.91	2.17	1.86	3.48	5.00
Unguided	0.81	1.71	0.66	0.02	0.53	1.00	0.66	0.96	1.22	1.83	2.73	2.43
Combined	1.49	3.64	0.89	0.01	0.67	0.96	0.65	1.10	1.42	1.84	3.02	4.22
Rock Sole												
Guided												
Unguided							0.01	0.01	0.01			
Combined							0.01	0.01	0.01			
Pacific Halibut												
Guided	1.89	4.97	1.17		2.43	0.29	0.40	1.91	2.17	1.86	3.48	5.00
Unguided	0.81	1.71	0.66	0.02	0.53	1.00	0.66	0.96	1.22	1.83	2.73	2.43
Combined	1.49	3.64	0.89	0.01	0.67	0.96	0.65	1.10	1.42	1.84	3.02	4.22
Sablefish												
Guided												
Unguided								0.02	0.03			
Combined								0.01	0.02			
Starry Flounder												
Guided												
Unguided												
Combined												
Unknown Groundfish												
Guided												
Unguided							0.04	0.02	0.01			
Combined							0.03	0.02	0.01			

Average number of rockfish harvested per boat trip in May 2021 by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided	n/a	n/a	n/a				0.69	0.69	0.71	1.67		0.71
Unguided				0.04	0.11	0.09	0.92	0.86	0.09	0.50	0.64	0.56
Combined				0.03	0.11	0.08	0.88	0.86	0.27	0.79	0.47	0.60
Boccacio Rockfish												
Guided	n/a	n/a	n/a								0.25	0.14
Unguided												
Combined											0.07	0.04
Canary Rockfish												
Guided	n/a	n/a	n/a									
Unguided									0.01	0.06		0.02
Combined									0.01	0.04		0.02
China Rockfish												
Guided	n/a	n/a	n/a				0.03	0.03	0.03		0.25	0.14
Unguided								0.14				
Combined								0.14	0.01		0.07	0.04
Copper Rockfish												
Guided	n/a	n/a	n/a				0.06	0.06	0.03			
Unguided						0.01			0.03			
Combined						0.01	0.04		0.03			
Dusky Rockfish												
Guided	n/a	n/a	n/a									
Unguided												
Combined												
Quillback Rockfish												
Guided	n/a	n/a	n/a		0.02	0.09	0.28	0.28	0.26	0.33	0.25	0.29
Unguided	0.09	0.09	0.17	0.01	0.10	0.19	0.25	0.14	0.11	0.06	0.36	0.22
Combined	0.07	0.07	0.14	0.01	0.10	0.18	0.28	0.14	0.15	0.13	0.33	0.24
Redband Rockfish												
Guided	n/a	n/a	n/a									
Unguided												
Combined												
Redstripe Rockfish												
Guided	n/a	n/a	n/a									
Unguided												
Combined												
Tiger Rockfish												
Guided	n/a	n/a	n/a									
Unguided					0.02	0.01			0.02		0.05	0.02
Combined					0.02	0.01			0.02		0.03	0.02
Widow Rockfish												
Guided	n/a	n/a	n/a									
Unguided						0.01						
Combined						0.01						
Yelloweye Rockfish												
Guided	n/a	n/a	n/a				0.03	0.03				
Unguided												
Combined							0.04					
Yellowtail Rockfish												
Guided	n/a	n/a	n/a								0.25	0.14
Unguided						0.03			0.01	0.06	0.05	0.05
Combined						0.02			0.01	0.04	0.10	0.07
Unknown rockfish												
Guided	n/a	n/a	n/a								1.13	0.64
Unguided				0.04		0.09	0.13		0.25			
Combined				0.03		0.08	0.12		0.18		0.30	0.16

Average number of rockfish harvested per boat trip, 1-18 June 2021, by trip type and subarea. Zeros hidden.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided	1.14	1.45	1.14		0.19	0.43	0.58	0.32	0.12	1.67		0.71
Unguided					0.05	0.35	0.22	0.38	0.46	0.17		0.08
Combined		0.94			0.28	0.36	0.38	0.36	0.35	0.67		0.30
Boccacio Rockfish												
Guided												
Unguided												
Combined												
Canary Rockfish												
Guided												
Unguided							0.09	0.03				
Combined							0.05	0.02				
China Rockfish												
Guided					0.02			0.02	0.03	0.17		0.07
Unguided					0.03			0.01	0.01			
Combined					0.07			0.01	0.02	0.06		0.03
Copper Rockfish												
Guided							0.08	0.09	0.09			
Unguided						0.05	0.06	0.09	0.10		0.21	0.12
Combined						0.04	0.07	0.09	0.10		0.15	0.08
Dusky Rockfish												
Guided												
Unguided							0.03	0.02	0.01			
Combined							0.02	0.01	0.01			
Quillback Rockfish												
Guided	0.29	0.18	0.29		0.20	1.21	0.38	0.72	0.97	0.33		0.14
Unguided	0.09	0.50	0.50		0.28	0.12	0.06	0.25	0.34			
Combined	0.23	0.29	0.43		0.26	0.32	0.20	0.42	0.54	0.11		0.05
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided										0.17		0.07
Unguided		0.17					0.03	0.01				
Combined		0.06					0.02	0.01		0.06		0.03
Widow Rockfish												
Guided												
Unguided						0.02						
Combined						0.01						
Yelloweye Rockfish												
Guided												
Unguided						0.02		0.01	0.01			
Combined						0.01		0.01	0.01			
Yellowtail Rockfish												
Guided												
Unguided					0.03			0.02	0.03			
Combined					0.02			0.01	0.02			
Unknown rockfish												
Guided											0.67	0.29
Unguided										0.08		0.04
Combined										0.06	0.20	0.13

Average number of rockfish harvested per boat trip, 19 June - 11 July 2021, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided	1.00	0.80	0.84		0.01	0.17	0.43	0.84	0.96		0.48	
Unguided	0.18	0.48			0.07	0.21	0.25	0.32	0.34	0.13	0.14	0.13
Combined	0.53	0.70			0.07	0.21	0.29	0.43	0.49	0.10	0.25	
Boccacio Rockfish												
Guided												
Unguided												
Combined												
Canary Rockfish												
Guided												
Unguided												
Combined												
China Rockfish												
Guided												
Unguided	0.14	0.35					0.07	0.05	0.05	0.04		0.01
Combined	0.08	0.11					0.06	0.04	0.04	0.03		
Copper Rockfish												
Guided		0.08	0.06				0.14	0.03			0.03	
Unguided		0.09				0.15						
Combined		0.08				0.14	0.03	0.01			0.01	
Dusky Rockfish												
Guided												
Unguided								0.03	0.04			
Combined								0.02	0.03			
Quillback Rockfish												
Guided	0.06	0.61	0.47		0.02	0.33	0.29	0.59	0.68	0.75	0.39	1.00
Unguided	0.18	0.30		0.01	0.05	0.15	0.29	0.51	0.60		0.30	0.21
Combined	0.13	0.51		0.00	0.07	0.16	0.29	0.53	0.62	0.19	0.33	0.63
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided											0.13	
Unguided												
Combined											0.04	
Tiger Rockfish												
Guided		0.02	0.01								0.03	
Unguided	0.09							0.02	0.02			
Combined	0.05	0.01						0.01	0.02		0.01	
Widow Rockfish												
Guided												
Unguided	0.05					0.02						
Combined	0.03					0.02						
Yelloweye Rockfish												
Guided												
Unguided												
Combined												
Yellowtail Rockfish												
Guided							0.57	0.13				
Unguided						0.03		0.04	0.05		0.02	0.01
Combined						0.03	0.11	0.06	0.04		0.01	
Unknown rockfish												
Guided		0.04	0.03					0.13	0.16		0.03	
Unguided			0.17			0.01		0.02	0.02			
Combined		0.03	0.14			0.01		0.04	0.06		0.01	

Average number of rockfish harvested per boat trip, 12-31 July 2021, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided	0.83	0.86	0.84		0.72		1.06	0.70	0.50	0.57	0.15	0.91
Unguided	0.14	0.28	0.11	0.01	0.12	0.17	0.21		0.44		0.69	0.51
Combined	0.30	0.64	0.11	0.00	0.23	0.15	0.35	0.25	0.45	0.19	0.56	0.91
Boccacio Rockfish												
Guided											0.08	
Unguided												
Combined											0.02	
Canary Rockfish												
Guided												
Unguided					0.04				0.01			
Combined					0.03				0.00			
China Rockfish												
Guided								0.04	0.06			
Unguided	0.10				0.07		0.01		0.02		0.10	0.08
Combined	0.07				0.06		0.01		0.02		0.08	
Copper Rockfish												
Guided					0.11	0.09	0.29	0.09				
Unguided					0.02	0.01	0.01		0.02			
Combined					0.04	0.02	0.06		0.02			
Dusky Rockfish												
Guided							0.06	0.02				
Unguided									0.01			
Combined							0.01		0.00			
Quillback Rockfish												
Guided		0.54	0.49		0.22		0.53	0.46	0.44	0.14	0.31	1.09
Unguided	0.29	0.22	0.11		0.04	0.14	0.07	0.14	0.28	0.14	0.08	0.09
Combined	0.22	0.42	0.11		0.07	0.13	0.14	0.13	0.31	0.14	0.13	1.09
Redband Rockfish												
Guided												
Unguided								0.14				
Combined								0.13				
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided		0.01	0.01		0.06			0.06	0.08			0.09
Unguided		0.02				0.03	0.01		0.02			
Combined		0.02			0.01	0.03	0.01		0.03			0.09
Widow Rockfish												
Guided												
Unguided							0.03		0.01			
Combined							0.03		0.00			
Yelloweye Rockfish												
Guided							0.06	0.02				
Unguided												
Combined							0.01					
Yellowtail Rockfish												
Guided												
Unguided				0.02	0.02	0.03	0.09		0.04		0.03	0.02
Combined				0.01	0.02	0.03	0.07		0.03		0.02	
Unknown rockfish												
Guided		0.49	0.45									
Unguided	0.10						0.03		0.03			
Combined	0.07	0.30					0.03		0.03			

Average number of rockfish harvested per boat trip in August 2021, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided	0.20	0.66	0.08		0.57	0.29	0.40	0.20	0.17			0.67
Unguided		0.46			0.11	0.14	0.45	0.51	0.58	0.50	0.09	0.22
Combined	0.12	0.58	0.04		0.15	0.15	0.44	0.46	0.49	0.36	0.06	0.44
Boccacio Rockfish												
Guided												
Unguided						0.01						
Combined						0.01						
Canary Rockfish												
Guided												
Unguided												
Combined												
China Rockfish												
Guided												
Unguided							0.01	0.03	0.04	0.06	0.03	0.04
Combined							0.01	0.02	0.03	0.04	0.02	
Copper Rockfish												
Guided	0.02	0.06						0.11	0.13			
Unguided		0.21					0.01	0.03	0.04		0.03	0.02
Combined	0.01	0.12					0.01	0.04	0.06		0.02	
Dusky Rockfish												
Guided												
Unguided						0.02		0.02	0.04			
Combined						0.02		0.02	0.03			
Quillback Rockfish												
Guided	0.22	0.46	0.04		0.57	0.29		0.37	0.43	0.71	0.10	
Unguided	0.04	0.38	0.14		0.11	0.04	0.30	0.26	0.23	0.11	0.15	0.19
Combined	0.15	0.42	0.09		0.15	0.05	0.28	0.27	0.27	0.28	0.13	0.33
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided								0.03	0.03	0.14		0.33
Unguided							0.02	0.01				
Combined							0.02	0.01	0.01	0.04		0.22
Widow Rockfish												
Guided												
Unguided								0.01	0.01			
Combined								0.00	0.01			
Yelloweye Rockfish												
Guided												
Unguided												
Combined												
Yellowtail Rockfish												
Guided			0.04									
Unguided	0.07	0.04	0.03			0.03		0.01	0.01		0.03	0.02
Combined	0.03	0.02	0.04			0.03		0.00	0.01		0.02	
Unknown rockfish												
Guided							0.40	0.06				
Unguided					0.01	0.05	0.04	0.04	0.04			
Combined					0.01	0.04	0.06	0.04	0.03			

APPENDIX 3: COUNTS OF ACTIVELY FISHING SPORT VESSELS IN EACH SUBAREA, BY DATE

Counts of actively fishing sport vessels in each subarea, by date, 2021, along with numbers that were estimated to be lodge boats. Weekend and holiday flights are shaded. * Flight scheduled for July 31, meant to represent weekend effort during July, was delayed until 1 August due to heavy fog.

Month	Date	Subarea												Area 3
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Lodge Boats
May	3 May 2021	0	0	0	2	4	4	0	0	1	1	0	0	0
	5 May 2021	0	0	0	0	3	4	0	0	8	0	1	0	0
	8 May 2021	1	1	0	5	0	4	1	1	1	0	0	0	0
	13 May 2021	0	1	3	1	4	3	1	0	2	1	1	0	0
	16 May 2021	3	9	1	0	9	5	2	2	17	3	1	2	0
	21 May 2021	5	6	3	1	11	9	0	0	16	0	3	0	0
	24 May 2021	1	3	0	6	8	8	3	0	9	1	2	1	0
	27 May 2021	2	0	8	1	3	6	1	0	7	2	0	0	5
June	3 Jun 2021	6	5	4	0	1	9	1	0	5	0	0	0	5
	5 Jun 2021	0	5	5	0	1	5	10	0	3	0	0	0	5
	10 Jun 2021	1	8	5	1	5	4	6	0	8	1	1	0	3
	13 Jun 2021	5	18	6	2	9	8	8	3	19	12	1	2	3
	18 Jun 2021	4	7	5	0	1	4	3	0	16	7	0	0	7
	22 Jun 2021	10	0	0	2	1	2	0	0	5	2	3	0	5
	26 Jun 2021	3	14	2	0	5	13	8	0	10	8	0	3	4
	30 Jun 2021	10	5	1	3	0	3	1	0	4	0	3	0	3
July	1 Jul 2021	20	12	2	7	2	19	2	0	3	0	0	0	3
	6 Jul 2021	8	19	2	1	0	9	3	0	10	0	1	0	7
	10 Jul 2021	7	29	6	0	0	21	1	0	15	13	7	5	7
	14 Jul 2021	7	22	4	2	3	12	2	0	30	8	1	2	4
	17 Jul 2021	5	13	3	1	5	16	9	0	40	15	1	3	4
	21 Jul 2021	3	25	9	1	15	25	3	0	9	1	0	1	7
	26 Jul 2021	5	9	4	4	2	10	22	1	23	0	1	0	0
	1 Aug 2021 *	6	15	10	4	3	28	12	3	28	15	1	2	4
August	2 Aug 2021	6	15	7	4	2	12	11	1	16	22	0	7	2
	6 Aug 2021	3	6	15	2	5	22	11	2	17	2	0	0	3
	12 Aug 2021	10	5	8	1	6	7	12	3	8	5	1	0	4
	15 Aug 2021	2	4	12	1	7	4	2	2	9	0	0	0	7
	19 Aug 2021	6	6	13	1	1	10	6	0	11	10	3	0	6
	21 Aug 2021	16	4	10	4	3	14	4	0	11	0	3	0	6
	25 Aug 2021	0	1	1	0	2	5	1	0	1	0	0	0	0
	30 Aug 2021	6	3	6	1	2	5	2	0	3	1	3	1	0

APPENDIX 4: RETAINED AND RELEASED FISH (LODGE REPORTS AND NON-LODGE CREEL ESTIMATES) BY MONTH, SUBAREA AND SPECIES.

Estimated numbers of Chinook Salmon retained by lodge and non-lodge anglers, by month, 2021.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	56.3	100.9	0	7.8	228.3	251.4	13.3	8.5	722.3	29.0	32.8	10.5	1,461
Lodge Report	42.6	76.4	0	0	0	0	0	0	0	0	0	0	119
MAY RETAINED	99.0	177.2	0	7.8	228.3	251.4	13.3	8.5	722.3	29.0	32.8	10.5	1,580
SE	32.1	58.2	0	2.6	34.4	34.2	4.5	5.9	53.2	13.8	10.0	3.8	100
June													
Creel Estimate	102.1	425.0	90.9	0.1	114.1	141.3	253.8	13.9	425.6	109.5	12.8	7.1	1,696
Lodge Report	22.8	94.9	20.3	0	0	0	0	0	0	0	0	0	138
JUNE RETAINED	125.0	519.8	111.2	0.1	114.1	141.3	253.8	13.9	425.6	109.5	12.8	7.1	1,834
SE	23.5	59.5	22.1	0.1	17.3	21.4	22.3	1.2	37.5	28.5	3.3	1.7	90
July													
Creel Estimate	147.3	902.1	30.6	0.3	150.8	239.1	191.7	5.1	621.0	289.0	22.2	21.8	2,621
Lodge Report	24.0	147.0	5.0	0	0	0	0	0	0	0	0	0	176
JULY RETAINED	171.3	1049.1	35.5	0.3	150.8	239.1	191.7	5.1	621.0	289.0	22.2	21.8	2,797
SE	29.9	65.2	12.3	0.3	22.7	49.1	20.6	3.4	65.1	92.5	3.6	12.4	147
August													
Creel Estimate	124.5	142.3	54.9	0	58.8	94.5	193.4	32.9	285.1	143.7	10.5	0	1,141
Lodge Report	12.0	13.7	5.3	0	0	0	0	0	0	0	0	0	31
AUGUST RETAINED	136.4	156.1	60.2	0	58.8	94.5	193.4	32.9	285.1	143.7	10.5	0	1,172
SE	19.8	23.2	18.4	0	10.4	18.0	28.4	3.0	33.8	54.0	5.1	0	81
TOTAL													
Creel Estimate	430	1,570	176	8	552	726	652	60	2,054	571	78	39	6,919
Lodge Report	101	332	31	0	0	0	0	0	0	0	0	0	464
TOTAL RETAINED	532	1,902	207	8	552	726	652	60	2,054	571	78	39	7,383
SE	54	108	31	3	46	66	42	8	98	112	12	13	215

Estimated numbers of Chinook Salmon released by lodge and non-lodge anglers, by month, 2021.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	5.6	10.1	0	0.9	153.4	60.2	4.4	0	239.5	0	5.8	1.1	481
Lodge Report	5.7	10.3	0	0	0	0	0	0	0	0	0	0	16
MAY RELEASED	11.4	20.4	0	0.9	153.4	60.2	4.4	0	239.5	0	5.8	1.1	497
SE	4.0	7.3	0	0.6	31.5	15.2	3.4	0	33.6	0	3.2	0.8	49
June													
Creel Estimate	762.0	1005.5	80.2	0.1	77.7	173.8	258.8	9.3	710.3	203.5	160.1	40.3	3,482
Lodge Report	70.1	92.5	7.4	0	0	0	0	0	0	0	0	0	170
JUNE RELEASED	832.1	1098.0	87.6	0.1	77.7	173.8	258.8	9.3	710.3	203.5	160.1	40.3	3,652
SE	98.5	153.9	40.2	0.1	21.2	38.9	38.0	1.4	72.7	44.8	29.0	27.4	217
July													
Creel Estimate	617.6	980.1	249.3	0	72.4	207.8	226.4	7.1	829.6	150.7	149.4	39.9	3,530
Lodge Report	54.8	87.0	22.1	0	0	0	0	0	0	0	0	0	164
JULY RELEASED	672.5	1067.1	271.4	0	72.4	207.8	226.4	7.1	829.6	150.7	149.4	39.9	3,694
SE	76.8	150.0	169.3	0	19.8	46.1	45.4	6.0	81.7	40.4	24.0	33.9	268
August													
Creel Estimate	172.0	104.4	824.1	0	49.3	116.8	144.0	19.1	138.4	215.6	6.6	0	1,790
Lodge Report	1.4	0.9	6.7	0	0	0	0	0	0	0	0	0	9
AUGUST RELEASED	173.4	105.2	830.9	0	49.3	116.8	144.0	19.1	138.4	215.6	6.6	0	1,799
SE	30.0	25.8	147.6	0	13.0	34.4	33.8	3.8	46.4	109.5	2.6	0	200
TOTAL													
Creel Estimate	1,557	2,100	1,154	1	353	559	634	36	1,918	570	322	81	9,284
Lodge Report	132	191	36	0	0	0	0	0	0	0	0	0	359
TOTAL RELEASED	1,689	2,291	1,190	1	353	559	634	36	1,918	570	322	81	9,643
SE	128	217	228	1	45	71	68	7	124	125	38	44	402

Estimated numbers of Coho Salmon retained by lodge and non-lodge anglers, by month, 2021.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	904.3	1114.6	8.9	0	1.8	71.5	38.6	0.2	351.7	81.9	89.1	32.1	2,695
Lodge Report	153.2	188.8	1.5	0	0	0	0	0	0	0	0	0	344
JUNE RETAINED	1057.4	1303.5	10.4	0	1.8	71.5	38.6	0.2	351.7	81.9	89.1	32.1	3,038
SE	112.4	99.2	6.5	0	0.8	15.2	9.5	0.1	41.0	26.2	13.0	22.6	161
July													
Creel Estimate	1602.1	5219.3	414.5	0.3	973.2	929.3	1538.1	50.7	4176.5	1170.2	209.9	175.3	16,460
Lodge Report	271.4	884.2	70.2	0	0	0	0	0	0	0	0	0	1,226
JULY RETAINED	1873.5	6103.5	484.7	0.3	973.2	929.3	1538.1	50.7	4176.5	1170.2	209.9	175.3	17,685
SE	154.5	221.3	132.1	0.3	84.6	102.9	105.1	17.4	173.9	155.0	17.3	51.6	420
August													
Creel Estimate	2529.4	1926.3	3618.3	0	457.3	1167.8	1888.7	316.8	2796.6	994.2	281.6	57.9	16,035
Lodge Report	428.5	326.3	612.9	0	0	0	0	0	0	0	0	0	1,368
AUGUST RETAINED	2957.8	2252.6	4231.2	0	457.3	1167.8	1888.7	316.8	2796.6	994.2	281.6	57.9	17,403
SE	120.0	152.3	233.4	0	49.2	135.1	132.0	13.9	158.6	195.8	39.9	50.0	445
TOTAL													
Creel Estimate	5,036	8,260	4,042	0	1,432	2,169	3,465	368	7,325	2,246	581	265	35,189
Lodge Report	853	1,399	685	0	0	0	0	0	0	0	0	0	2,937
TOTAL RETAINED	5,889	9,660	4,726	0	1,432	2,169	3,465	368	7,325	2,246	581	265	38,126
SE	226	286	268	0	98	171	169	22	239	251	45	75	633

Estimated numbers of Coho Salmon released by lodge and non-lodge anglers, by month, 2021.

Month / Catch Type	Subarea												TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	96.9	106.6	0	0	0	3.8	9.1	0.2	50.0	31.7	12.8	0	311
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	96.9	106.6	0	0	0	3.8	9.1	0.2	50.0	31.7	12.8	0	311
SE	25.6	50.0	0	0	0	1.9	3.3	0.1	22.0	14.8	4.7	0	62
July													
Creel Estimate	344.1	639.1	30.6	0	158.9	202.5	357.7	1.0	715.6	316.8	47.3	0	2,814
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	344.1	639.1	30.6	0	158.9	202.5	357.7	1.0	715.6	316.8	47.3	0	2,814
SE	88.0	105.3	20.9	0	44.6	50.7	73.6	1.0	90.4	109.3	12.5	0	223
August													
Creel Estimate	241.6	256.2	1467.7	0	74.0	222.4	288.0	41.7	318.7	95.8	31.6	0	3,038
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	241.6	256.2	1467.7	0	74.0	222.4	288.0	41.7	318.7	95.8	31.6	0	3,038
SE	49.7	74.5	235.0	0	20.0	61.0	78.1	6.8	67.3	42.3	10.1	0	283
TOTAL													
Creel Estimate	683	1,002	1,498	0	233	429	655	43	1,084	444	92	0	6,163
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	683	1,002	1,498	0	233	429	655	43	1,084	444	92	0	6,163
SE	104	138	236	0	49	79	107	7	115	118	17	0	365

Estimated numbers of Pink Salmon retained by lodge and non-lodge anglers, by month, 2021.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	72.7	45.3	0	0.2	0	5.6	0	0	17.6	5.6	3.7	0	151
Lodge Report	0.7	0.4	0	0	0	0	0	0	0	0	0	0	1
JUNE RETAINED	73.3	45.7	0	0.2	0	5.6	0	0	17.6	5.6	3.7	0	152
SE	15.5	12.5	0	0.1	0	3.1	0	0	7.9	3.5	1.2	0	22
July													
Creel Estimate	131.8	201.7	76.4	0.2	28.2	127.9	75.6	0	194.5	33.6	6.8	21.8	898
Lodge Report	1.2	1.8	0.7	0	0	0	0	0	0	0	0	0	4
JULY RETAINED	133.0	203.5	77.1	0.2	28.2	127.9	75.6	0	194.5	33.6	6.8	21.8	902
SE	29.4	31.1	60.6	0.1	7.5	34.7	15.1	0	25.1	15.0	1.5	12.4	90
August													
Creel Estimate	109.8	175.5	188.4	0	13.3	72.3	57.6	14.4	155.1	0	14.5	0	801
Lodge Report	1.0	1.6	1.7	0	0	0	0	0	0	0	0	0	4
AUGUST RETAINED	110.8	177.1	190.1	0	13.3	72.3	57.6	14.4	155.1	0	14.5	0	805
SE	19.0	39.9	48.2	0	6.3	20.1	15.3	2.6	35.6	0	4.7	0	79
TOTAL													
Creel Estimate	314	423	265	0	41	206	133	14	367	39	25	22	1,850
Lodge Report	3	4	2	0	0	0	0	0	0	0	0	0	9
TOTAL RETAINED	317	426	267	0	41	206	133	14	367	39	25	22	1,859
SE	38	52	77	0	10	40	22	3	44	15	5	12	121

Estimated numbers of Pink Salmon released by lodge and non-lodge anglers, by month, 2021.

Month / Catch Type	Subarea												TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	56.5	106.6	8.9	0	0.4	5.6	0	0	7.4	0	1.1	0	186
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	56.5	106.6	8.9	0	0.4	5.6	0	0	7.4	0	1.1	0	186
SE	29.7	41.6	6.5	0	0.3	3.1	0	0	4.4	0	0.8	0	52
July													
Creel Estimate	221.8	1421.5	200.5	0	153.2	82.7	218.7	5.1	454.7	225.8	27.0	65.5	3,076
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	221.8	1421.5	200.5	0	153.2	82.7	218.7	5.1	454.7	225.8	27.0	65.5	3,076
SE	74.9	221.3	89.0	0	38.9	22.2	36.1	5.1	74.6	58.4	6.2	46.4	278
August													
Creel Estimate	640.6	327.4	1020.3	0	19.0	328.1	201.6	65.8	792.5	395.3	51.3	0	3,842
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	640.6	327.4	1020.3	0	19.0	328.1	201.6	65.8	792.5	395.3	51.3	0	3,842
SE	77.2	66.3	189.8	0	6.7	88.6	39.8	8.1	113.5	95.5	15.7	0	280
TOTAL													
Creel Estimate	919	1,855	1,230	0	173	416	420	71	1,254	621	79	66	7,105
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	919	1,855	1,230	0	173	416	420	71	1,254	621	79	66	7,105
SE	112	235	210	0	39	91	54	10	136	112	17	46	397

Estimated numbers of Chum Salmon retained by lodge and non-lodge anglers, by month, 2021.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	4.0	0	0	0	0	0	0	0	0	0	1.1	0	5.1
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	4.0	0	0	0	0	0	0	0	0	0	1.1	0	5.1
SE	2.9	0	0	0	0	0	0	0	0	0	0.8	0	3.7
July													
Creel Estimate	3.8	0	0	0	0	0	0	0	0	0	1.0	0	4.8
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	3.8	0	0	0	0	0	0	0	0	0	1.0	0	4.8
SE	2.7	0	0	0	0	0	0	0	0	0	0.7	0	3.4
August													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL													
Creel Estimate	8	0	0	0	0	0	0	0	0	0	2	0	10
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	8	0	0	0	0	0	0	0	0	0	2	0	10
SE	4	0	0	0	0	0	0	0	0	0	1	0	4

Estimated numbers of Chum Salmon released by lodge and non-lodge anglers, by month, 2021.

Month / Catch Type	Subarea												TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
August													
Creel Estimate	0	4.7	0	0	0	0	0	0	0	0	0	0	5
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	4.7	0	0	0	0	0	0	0	0	0	0	5
SE	0	3.4	0	0	0	0	0	0	0	0	0	0	3
TOTAL													
Creel Estimate	0	5	0	0	0	0	0	0	0	0	0	0	5
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	0	5	0	0	0	0	0	0	0	0	0	0	5
SE	0	3	0	0	0	0	0	0	0	0	0	0	3

Estimated numbers of Sockeye Salmon retained by lodge and non-lodge anglers, by month, 2021.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	4.0	0	0	0	0	0	0	0	0	0	0	4
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	4.0	0	0	0	0	0	0	0	0	0	0	4
SE	0	3.4	0	0	0	0	0	0	0	0	0	0	3
August													
Creel Estimate	0	0	0	0	0	0	12.3	0.8	0	0	0	5.3	18
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	0	0	0	0	0	12.3	0.8	0	0	0	5.3	18
SE	0	0	0	0	0	0	6.7	0.4	0	0	0	4.5	8
TOTAL													
Creel Estimate	0	4	0	0	0	0	12	1	0	0	0	5	22
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	0	4	0	0	0	0	12	1	0	0	0	5	22
SE	0	3	0	0	0	0	7	0	0	0	0	5	9

Estimated numbers of Sockeye Salmon released by lodge and non-lodge anglers, by month, 2021.

Month / Catch Type	Subarea												TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
August													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0

Estimated numbers of unknown salmon retained by lodge and non-lodge anglers, by month, 2021.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	3.6	0	0	0	0	0	0	0	4
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	3.6	0	0	0	0	0	0	0	4
SE	0	0	0	0	2.6	0	0	0	0	0	0	0	3
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	4.5	0	0	0	0	0	0	5
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	0	0	0	0	4.5	0	0	0	0	0	0	5
SE	0	0	0	0	0	3.6	0	0	0	0	0	0	4
August													
Creel Estimate	0	0	0	0	0	0	12.3	0.8	0	0	0	0	13
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	0	0	0	0	0	12.3	0.8	0	0	0	0	13
SE	0	0	0	0	0	0	9.0	0.6	0	0	0	0	9
TOTAL													
Creel Estimate	0	0	0	0	4	5	12	1	0	0	0	0	21
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	0	0	0	0	4	5	12	1	0	0	0	0	21
SE	0	0	0	0	3	4	9	1	0	0	0	0	10

Estimated numbers of unknown salmon released by lodge and non-lodge anglers, by month, 2021.

Month / Catch Type	Subarea												TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	0	0	0	1.9	0	0	0	0	0	0	2
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	1.9	0	0	0	0	0	0	2
SE	0	0	0	0	0	1.4	0	0	0	0	0	0	1
July													
Creel Estimate	0	0	0	0	0	2.0	0	0	0	0	0	0	2
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	2.0	0	0	0	0	0	0	2
SE	0	0	0	0	0	1.6	0	0	0	0	0	0	2
August													
Creel Estimate	0	0	0	0	0	0	8.2	0.6	0	0	0	0	9
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	0	0	8.2	0.6	0	0	0	0	9
SE	0	0	0	0	0	0	6.0	0.4	0	0	0	0	6
TOTAL													
Creel Estimate	0	0	0	0	0	4	8	1	0	0	0	0	13
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	0	0	0	0	0	4	8	1	0	0	0	0	13
SE	0	0	0	0	0	2	6	0	0	0	0	0	6

Estimated numbers of Pacific Halibut retained by lodge and non-lodge anglers, by month, 2021.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	101.4	181.5	11.8	2.6	42.8	269.1	59.9	22.7	646.1	77.3	48.2	23.5	1,487
Lodge Report	4.1	7.4	0.5	0	0	0	0	0	0	0	0	0	12
MAY RETAINED	105.5	188.9	12.3	2.6	42.8	269.1	59.9	22.7	646.1	77.3	48.2	23.5	1,499
SE	38.1	69.0	10.9	1.4	12.4	36.6	18.0	14.7	64.2	17.7	10.3	5.2	114
June													
Creel Estimate	292.5	925.4	28.4	0.4	92.5	486.6	271.5	15.4	890.1	288.7	146.0	90.0	3,527
Lodge Report	46.5	147.0	4.5	0	0	0	0	0	0	0	0	0	198
JUNE RETAINED	339.0	1072.4	33.0	0.4	92.5	486.6	271.5	15.4	890.1	288.7	146.0	90.0	3,725
SE	54.2	91.7	12.9	0.3	20.0	39.6	38.1	2.0	73.1	57.6	11.8	16.4	155
July													
Creel Estimate	512.2	1919.6	182.3	0.4	161.6	694.8	361.0	13.2	1348.8	565.1	152.7	293.5	6,205
Lodge Report	61.3	229.8	21.8	0	0	0	0	0	0	0	0	0	313
JULY RETAINED	573.5	2149.4	204.2	0.4	161.6	694.8	361.0	13.2	1348.8	565.1	152.7	293.5	6,518
SE	77.3	128.4	63.9	0.3	30.3	68.6	50.3	7.8	83.5	86.3	11.0	43.1	226
August													
Creel Estimate	399.0	1020.1	368.9	1.1	121.4	617.3	234.6	70.5	830.2	551.0	214.5	200.0	4,629
Lodge Report	66.9	171.2	61.9	0	0	0	0	0	0	0	0	0	300
AUGUST RETAINED	465.9	1191.2	430.8	1.1	121.4	617.3	234.6	70.5	830.2	551.0	214.5	200.0	4,929
SE	53.5	85.1	71.9	0.6	21.4	71.3	36.3	5.9	80.1	103.2	23.8	28.0	201
TOTAL													
Creel Estimate	1,305	4,047	592	4	418	2,068	927	122	3,715	1,482	561	607	15,848
Lodge Report	179	555	89	0	0	0	0	0	0	0	0	0	823
TOTAL RETAINED	1,484	4,602	680	4	418	2,068	927	122	3,715	1,482	561	607	16,671
SE	115	192	98	2	44	113	75	18	151	147	31	54	359

Estimated numbers of Pacific Halibut released by lodge and non-lodge anglers, by month, 2021.

Month / Catch Type	Subarea												TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	
May													
Creel Estimate	236.6	423.6	0	0	42.8	49.6	6.7	2.8	344.8	38.7	38.6	13.0	1,197
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	236.6	423.6	0	0	42.8	49.6	6.7	2.8	344.8	38.7	38.6	13.0	1,197
SE	161.0	291.4	0	0	18.9	15.2	3.8	2.8	77.6	15.4	13.7	4.7	343
June													
Creel Estimate	159.3	454.5	0	0	21.5	117.7	74.1	4.9	370.1	220.7	86.9	52.1	1,562
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	159.3	454.5	0	0	21.5	117.7	74.1	4.9	370.1	220.7	86.9	52.1	1,562
SE	58.6	92.4	0	0	10.0	22.5	18.3	1.2	60.2	75.9	16.4	20.5	152
July													
Creel Estimate	181.1	1270.3	129.9	0	147.2	294.0	98.1	5.1	490.7	245.0	92.2	234.6	3,188
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	181.1	1270.3	129.9	0	147.2	294.0	98.1	5.1	490.7	245.0	92.2	234.6	3,188
SE	55.3	222.9	83.6	0	53.0	67.0	27.6	5.1	75.4	92.7	15.1	55.7	292
August													
Creel Estimate	237.9	906.2	188.4	0	24.7	144.6	78.2	20.7	234.8	179.7	64.5	89.5	2,169
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	237.9	906.2	188.4	0	24.7	144.6	78.2	20.7	234.8	179.7	64.5	89.5	2,169
SE	70.6	161.9	68.4	0	10.2	38.9	28.8	3.7	47.7	103.4	14.3	32.3	229
TOTAL													
Creel Estimate	815	3,055	318	0	236	606	257	34	1,440	684	282	389	8,116
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	815	3,055	318	0	236	606	257	34	1,440	684	282	389	8,116
SE	193	412	108	0	58	82	44	7	133	159	30	68	528

Estimated numbers of Lingcod retained by lodge and non-lodge anglers, by month, 2021.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	78.9	141.2	0	0.9	14.3	92.1	73.2	65.3	439.2	144.9	63.7	33.6	1,147
Lodge Report	11.1	19.9	0	0	0	0	0	0	0	0	0	0	31
MAY RETAINED	90.0	161.1	0	0.9	14.3	92.1	73.2	65.3	439.2	144.9	63.7	33.6	1,178
SE	48.5	87.8	0	0.6	5.1	26.8	29.8	44.0	59.9	34.2	21.9	9.2	138
June													
Creel Estimate	120.6	433.0	8.9	0.2	21.9	103.5	182.3	13.1	601.8	271.6	47.0	38.0	1,842
Lodge Report	59.6	214.0	4.4	0	0	0	0	0	0	0	0	0	278
JUNE RETAINED	180.2	647.0	13.2	0.2	21.9	103.5	182.3	13.1	601.8	271.6	47.0	38.0	2,120
SE	45.3	74.6	6.5	0.1	7.7	25.1	40.7	2.3	82.3	51.3	10.0	22.8	142
July													
Creel Estimate	102.2	738.7	17.1	0.2	24.1	114.1	109.7	2.0	542.2	81.3	35.9	112.8	1,880
Lodge Report	51.5	371.9	8.6	0	0	0	0	0	0	0	0	0	432
JULY RETAINED	153.7	1110.6	25.7	0.2	24.1	114.1	109.7	2.0	542.2	81.3	35.9	112.8	2,312
SE	31.4	83.1	10.1	0.1	9.3	25.4	22.1	2.0	64.0	17.8	7.2	46.4	126
August													
Creel Estimate	146.4	360.6	149.1	0	11.4	77.9	82.3	28.7	318.7	179.7	17.1	57.9	1,430
Lodge Report	67.6	166.5	68.9	0	0	0	0	0	0	0	0	0	303
AUGUST RETAINED	214.0	527.1	218.0	0	11.4	77.9	82.3	28.7	318.7	179.7	17.1	57.9	1,733
SE	36.2	80.1	69.2	0	5.2	22.6	33.0	4.4	52.8	58.0	10.3	40.2	148
TOTAL													
Creel Estimate	448	1,673	175	1	72	387	448	109	1,902	677	164	242	6,299
Lodge Report	190	772	82	0	0	0	0	0	0	0	0	0	1,044
TOTAL RETAINED	638	2,446	257	1	72	387	448	109	1,902	677	164	242	7,343
SE	82	163	70	1	14	50	64	44	131	86	27	66	277

Estimated numbers of Lingcod released by lodge and non-lodge anglers, by month, 2021.

Month / Catch Type	Subarea												TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	
May													
Creel Estimate	0	0	0	0	7.1	28.3	26.6	19.9	43.6	12.1	28.9	7.2	174
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	7.1	28.3	26.6	19.9	43.6	12.1	28.9	7.2	174
SE	0	0	0	0	5.2	14.2	20.5	14.4	17.6	6.1	13.6	3.7	37
June													
Creel Estimate	8.1	38.0	0	0.2	0.7	5.2	0	0.9	55.0	37.9	6.3	1.6	154
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	8.1	38.0	0	0.2	0.7	5.2	0	0.9	55.0	37.9	6.3	1.6	154
SE	5.8	18.2	0	0.1	0.6	4.2	0	0.3	14.8	13.5	2.7	0.9	28
July													
Creel Estimate	7.6	49.3	0	0.2	0.8	0	5.4	0	42.5	26.7	2.9	0	136
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	7.6	49.3	0	0.2	0.8	0	5.4	0	42.5	26.7	2.9	0	136
SE	5.4	27.9	0	0.1	0.7	0	3.0	0	12.9	14.2	1.1	0	34
August													
Creel Estimate	22.0	147.1	0	0	22.8	61.2	0	1.7	12.6	0	3.9	0	271
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	22.0	147.1	0	0	22.8	61.2	0	1.7	12.6	0	3.9	0	271
SE	13.3	74.8	0	0	16.7	26.3	0	0.7	5.1	0	3.1	0	82
TOTAL													
Creel Estimate	38	234	0	0	31	95	32	22	154	77	42	9	734
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	38	234	0	0	31	95	32	22	154	77	42	9	734
SE	16	82	0	0	18	30	21	14	27	20	14	4	101

Estimated numbers of other groundfish retained by lodge and non-lodge anglers, by month, 2021. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	22.5	40.3	0	0	10.7	24.8	6.7	0	3.6	12.1	11.6	4.0	136
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	22.5	40.3	0	0	10.7	24.8	6.7	0	3.6	12.1	11.6	4.0	136
SE	11.0	19.9	0	0	5.8	7.7	5.1	0	2.6	8.6	7.3	2.6	28
June													
Creel Estimate	9.7	70.2	0	0.2	7.1	45.4	5.1	0.9	78.5	28.9	33.2	5.2	285
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	9.7	70.2	0	0.2	7.1	45.4	5.1	0.9	78.5	28.9	33.2	5.2	285
SE	5.5	33.9	0	0.1	5.7	12.1	2.2	0.4	20.3	12.6	7.8	4.6	0
July													
Creel Estimate	24.9	71.0	38.2	0.2	24.1	89.4	24.2	11.2	157.3	28.6	32.0	31.9	533
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	24.9	71.0	38.2	0.2	24.1	89.4	24.2	11.2	157.3	28.6	32.0	31.9	533
SE	10.8	18.1	27.8	0.1	11.0	22.0	7.5	10.2	34.7	12.7	6.8	19.4	61
August													
Creel Estimate	58.6	33.2	23.5	0	24.7	66.7	94.6	15.2	134.2	0	44.7	0	495
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	58.6	33.2	23.5	0	24.7	66.7	94.6	15.2	134.2	0	44.7	0	495
SE	20.3	12.9	12.7	0	8.8	20.8	28.0	2.8	31.6	0	11.9	0	56
TOTAL													
Creel Estimate	116	215	62	0	67	226	131	27	374	70	122	41	1,449
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	116	215	62	0	67	226	131	27	374	70	122	41	1,449
SE	26	45	31	0	16	33	29	11	51	20	17	20	99

Estimated numbers of other groundfish released by lodge and non-lodge anglers, by month, 2021. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.

Month / Catch Type	Subarea												TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	
May													
Creel Estimate	0	0	0	0	7.1	24.8	0	11.4	50.8	0	1.9	0.7	97
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	7.1	24.8	0	11.4	50.8	0	1.9	0.7	97
SE	0	0	0	0	3.7	6.8	0	8.0	16.5	0	1.4	0.5	20
June													
Creel Estimate	4.0	5.3	0	2.0	20.3	190.9	9.1	1.0	56.3	3.7	8.1	0.2	301
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	4.0	5.3	0	2.0	20.3	190.9	9.1	1.0	56.3	3.7	8.1	0.2	301
SE	2.9	3.8	0	1.5	8.6	106.5	3.1	0.4	17.3	3.2	2.8	0.2	45
July													
Creel Estimate	19.6	17.4	30.6	2.7	90.1	108.5	12.1	1.0	227.9	13.4	12.6	4.4	540
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	19.6	17.4	30.6	2.7	90.1	108.5	12.1	1.0	227.9	13.4	12.6	4.4	540
SE	9.9	8.6	21.3	1.4	50.6	23.3	5.1	1.0	62.7	7.6	3.2	3.1	88
August													
Creel Estimate	36.6	4.7	102.0	2.7	119.5	344.8	61.7	6.4	33.5	263.5	42.1	0	1,018
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	36.6	4.7	102.0	2.7	119.5	344.8	61.7	6.4	33.5	263.5	42.1	0	1,018
SE	13.3	3.4	40.0	1.3	41.5	73.3	22.7	1.7	12.6	97.2	15.1	0	139
TOTAL													
Creel Estimate	60	27	133	7	237	669	83	20	369	281	65	5	1,955
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	60	27	133	7	237	669	83	20	369	281	65	5	1,955
SE	17	10	45	2	66	132	23	8	68	98	16	3	198

Estimated numbers of rockfish retained by lodge and non-lodge anglers, by month, 2021. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

Month / Catch Type	Subarea												TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	
May													
Creel Estimate	5.6	10.1	11.8	7.8	74.9	123.9	75.5	22.7	301.2	58.0	79.1	23.5	794
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	5.6	10.1	11.8	7.8	74.9	123.9	75.5	22.7	301.2	58.0	79.1	23.5	794
SE	4.0	7.3	10.9	3.7	20.4	26.1	17.8	14.6	51.7	21.6	21.0	7.0	74
June													
Creel Estimate	142.2	524.8	49.9	0.2	62.0	253.8	141.8	9.0	544.7	122.0	37.9	15.3	1,904
Lodge Report	0	17.6	14.6	0	0	0	0	0	0	0	0	0	32
JUNE RETAINED	142.2	542.4	64.5	0.2	62.0	253.8	141.8	9.0	544.7	122.0	37.9	15.3	1,936
SE	29.2	79.1	36.4	0.1	19.0	39.7	23.7	1.5	60.6	42.8	5.9	9.2	0
July													
Creel Estimate	213.0	960.2	35.3	1.4	95.7	299.9	240.8	4.1	923.9	84.9	50.9	114.7	3,025
Lodge Report	34.8	506.3	15.7	0	0	0	0	0	0	0	0	0	557
JULY RETAINED	247.8	1466.5	51.0	1.4	95.7	299.9	240.8	4.1	923.9	84.9	50.9	114.7	3,582
SE	36.4	115.7	16.8	0.9	21.0	45.3	36.4	2.5	82.3	31.8	6.6	29.2	166
August													
Creel Estimate	84.2	317.9	70.6	0	55.0	189.1	300.4	55.8	540.9	215.6	17.1	47.4	1,894
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	84.2	317.9	70.6	0	55.0	189.1	300.4	55.8	540.9	215.6	17.1	47.4	1,894
SE	21.1	50.6	27.4	0	12.7	35.7	54.9	6.2	77.0	59.7	5.0	21.4	135
TOTAL													
Creel Estimate	445	1,813	168	9	288	867	758	92	2,311	480	185	201	7,617
Lodge Report	35	524	30	0	0	0	0	0	0	0	0	0	589
TOTAL RETAINED	480	2,337	198	9	288	867	758	92	2,311	480	185	201	8,206
SE	51	149	50	4	37	75	72	16	138	83	23	38	260

Estimated numbers of rockfish released by lodge and non-lodge anglers, by month, 2021. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

Month / Catch Type	Subarea												TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	
May													
Creel Estimate	16.9	30.3	0	13.0	278.3	180.6	108.7	17.0	635.2	60.4	110.0	29.6	1,480
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	16.9	30.3	0	13.0	278.3	180.6	108.7	17.0	635.2	60.4	110.0	29.6	1,480
SE	9.0	16.3	0	3.9	52.9	34.8	44.7	12.7	109.5	19.2	51.9	13.7	148
June													
Creel Estimate	237.1	326.2	43.6	0	63.7	230.6	174.9	17.0	832.6	166.0	46.5	9.1	2,147
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	237.1	326.2	43.6	0	63.7	230.6	174.9	17.0	832.6	166.0	46.5	9.1	2,147
SE	77.6	105.1	25.2	0	20.4	41.2	39.2	2.9	114.9	45.5	10.6	4.9	129
July													
Creel Estimate	158.6	339.8	21.9	0.3	118.6	436.2	315.4	1.0	1018.6	101.5	56.2	53.7	2,622
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	158.6	339.8	21.9	0.3	118.6	436.2	315.4	1.0	1018.6	101.5	56.2	53.7	2,622
SE	37.7	74.5	10.7	0.3	24.7	118.9	63.2	1.0	117.7	38.2	10.1	21.0	204
August													
Creel Estimate	212.3	180.3	329.7	0	70.2	394.8	283.9	76.8	826.0	179.7	25.0	5.3	2,584
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	212.3	180.3	329.7	0	70.2	394.8	283.9	76.8	826.0	179.7	25.0	5.3	2,584
SE	65.7	43.1	76.6	0	20.8	112.0	66.0	11.4	154.4	74.9	8.5	4.5	243
TOTAL													
Creel Estimate	625	877	395	13	531	1,242	883	112	3,312	508	238	98	8,833
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	625	877	395	13	531	1,242	883	112	3,312	508	238	98	8,833
SE	109	137	81	4	65	172	109	17	251	97	55	26	399

APPENDIX 5: ADIPOSE FIN-CLIP INCIDENCE

Incidence of fin clipping observed in biosamples, by species, month, and subarea, in 2021.

Incidence of fin clipping observed in biosamples, by month and subarea, in 2021. Excludes any interviews for which observers were not able to confirm adipose status, or for which biosampling was not a complete sample of fish kept.

Species		Subarea														
Month	Adipose	3I	3J	3K	3?	4A	4B	4C	4D	4E	4F	4G	4H	4L	4?	TOTAL
Chinook Salmon																
May	Clipped	0	0	0	0	2	1	4	1	0	3	0	0	0	0	11
	Not Clipped	1	0	0	0	7	21	17	1	1	62	1	10	0	0	121
	Percent Clipped	0%	-	-	-	22%	5%	19%	50%	0%	5%	0%	0%	-	-	8%
June	Clipped	0	6	0	0	0	0	6	7	0	4	0	0	0	0	23
	Not Clipped	10	13	4	0	0	21	21	55	0	56	8	9	0	0	197
	Percent Clipped	0%	32%	0%	-	-	0%	22%	11%	-	7%	0%	0%	-	-	10%
July	Clipped	0	1	0	0	0	0	1	0	0	0	0	1	0	0	3
	Not Clipped	2	5	0	0	0	5	7	7	0	12	0	0	0	0	38
	Percent Clipped	0%	17%	-	-	-	0%	13%	0%	-	0%	-	100%	-	-	7%
August	Clipped	1	0	0	0	0	1	0	0	0	0	0	1	0	0	3
	Not Clipped	1	0	0	0	0	3	3	1	0	3	1	0	0	0	12
	Percent Clipped	50%	-	-	-	-	25%	0%	0%	-	0%	0%	100%	-	-	20%
Total	Clipped	1	7	0	0	2	2	11	8	0	7	0	2	0	0	40
	Not Clipped	14	18	4	0	7	50	48	64	1	133	10	19	0	0	368
	Percent Clipped	7%	28%	0%	-	22%	4%	19%	11%	0%	5%	0%	10%	-	-	9.8%
Coho Salmon																
May	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Percent Clipped	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
June	Clipped	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1
	Not Clipped	0	34	0	0	0	0	1	0	0	6	0	4	0	0	45
	Percent Clipped	-	0%	-	-	-	-	0%	-	-	0%	-	20%	-	-	2.2%
July	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	10	28	0	0	0	14	26	55	0	105	5	16	0	0	259
	Percent Clipped	0%	0%	-	-	-	0%	0%	0%	-	0%	0%	0%	-	-	0.0%
August	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	31	0	24	0	0	26	22	47	0	29	7	7	0	0	193
	Percent Clipped	0%	-	0%	-	-	0%	0%	0%	-	0%	0%	0%	-	-	0.0%
Total	Clipped	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1
	Not Clipped	41	62	24	0	0	40	49	102	0	140	12	27	0	0	497
	Percent Clipped	0%	0%	0%	-	-	0%	0%	0%	-	0%	0%	4%	-	-	0.2%